```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -5.45400e+03-3.35846e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.37264e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +5.45400e+03-3.35846e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.69514e+03+3.36277e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 495,496...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 495,496...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +8.75298e+03+1.41720e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +6.47122e+01-1.16028e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.36000e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -6.47122e+01-1.16028e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +8.75298e+03-1.41720e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 497,496...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
4.487 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 497,496...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.39693e+02-1.34499e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +6.97483e+02+1.73003e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -7.16691e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -6.97483e+02+1.73003e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.39693e+02+1.34499e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 498,496...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.22326e+02+5.43388e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -6.00587e+02+1.41037e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -6.57366e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +6.00587e+02+1.41037e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.22326e+02-5.43388e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 500,496...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.71734e+03+2.43967e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.05800e+02+1.49697e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +8.86222e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.05800e+02+1.49697e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.71734e+03-2.43967e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 501,496...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.26702e+02-4.67700e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.19943e+03+1.16737e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.57571e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.19943e+03+1.16737e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.26702e+02+4.67700e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 502,496...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.26737e+02-1.16648e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.52116e+03+1.06327e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.17962e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.52116e+03+1.06327e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.26737e+02+1.16648e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 503,496...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.78883e+03-9.82833e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +5.29358e+03+5.37610e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.62732e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -5.29358e+03+5.37610e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.78883e+03+9.82833e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 504,496...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.88140e+02-4.78604e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.44058e+03+1.62844e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.53697e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.44058e+03+1.62844e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.88140e+02+4.78604e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 344,497...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.70962e+03+1.94271e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.56754e+02+3.46756e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.09012e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.56754e+02+3.46756e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.70962e+03-1.94271e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 346,497...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.24138e+03-1.96597e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.05255e+03-3.76502e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.44949e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.05255e+03-3.76502e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.24138e+03+1.96597e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 458,497...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.71209e+03-1.93448e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.64814e+03+9.28158e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +7.61150e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.64814e+03+9.28158e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.71209e+03+1.93448e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 493,497...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.66624e+03+1.70967e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.09336e+03-4.61145e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.70845e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.09336e+03-4.61145e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.66624e+03-1.70967e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 494,497...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.35151e+02-4.50150e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.27633e+03+2.72439e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.11915e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -3.27633e+03+2.72439e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.35151e+02+4.50150e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 495,497...
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.40843e+01+7.84813e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.01056e+03-1.94980e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.43736e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.01056e+03-1.94980e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.40843e+01-7.84813e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 496,497...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.39693e+02-1.34499e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +6.97483e+02+1.73003e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -7.16691e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -6.97483e+02+1.73003e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.39693e+02+1.34499e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 498,497...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.84030e+03+6.05262e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.47446e+02+7.04236e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.41652e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.47446e+02+7.04236e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.84030e+03-6.05262e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 499,497...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.25818e+03-1.84620e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.21035e+02+1.29674e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.42347e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.21035e+02+1.29674e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.25818e+03+1.84620e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 500,497...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.61114e+03+8.50885e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.83211e+02-2.41281e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.35508e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -3.83211e+02-2.41281e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.61114e+03-8.50885e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 501,497...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.72305e+02-1.55504e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.55073e+03-2.01213e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.38928e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.55073e+03-2.01213e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.72305e+02+1.55504e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 502,497...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.16227e+03-4.67055e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.92125e+03-1.01938e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.26298e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.92125e+03-1.01938e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.16227e+03+4.67055e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 503,497...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.81966e+03+1.18960e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.99296e+03+9.11631e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.83365e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.99296e+03+9.11631e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.81966e+03-1.18960e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 19,498...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.50799e+03-1.90478e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +9.11681e+02+2.70775e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.51322e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -9.11681e+02+2.70775e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.50799e+03+1.90478e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 25,498...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.95963e+02-1.79626e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.64651e+02-8.52608e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.17899e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +7.64651e+02-8.52608e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.95963e+02+1.79626e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 28,498...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.02953e+03+2.87450e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.29062e+03+1.83315e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.82099e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.29062e+03+1.83315e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.02953e+03-2.87450e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 44,498...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.33973e+02-2.92836e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -9.44065e+01-8.74022e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.18870e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +9.44065e+01-8.74022e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.33973e+02+2.92836e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 488,498...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.39654e+03-5.94765e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.00238e+03-1.73994e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.33717e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.00238e+03-1.73994e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.39654e+03+5.94765e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 493,498...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.33170e+03-6.15619e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +7.60233e+03+3.47301e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.14777e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -7.60233e+03+3.47301e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.33170e+03+6.15619e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 494,498...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.51613e+03-4.38088e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.53482e+02+1.08407e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +9.78450e+02 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.53482e+02+1.08407e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.51613e+03+4.38088e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 495,498...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.63332e+02+1.66704e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -4.75149e+02+2.17498e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.73527e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +4.75149e+02+2.17498e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.63332e+02-1.66704e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 496,498...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.22326e+02+5.43388e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -6.00587e+02+1.41037e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -6.57366e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +6.00587e+02+1.41037e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.22326e+02-5.43388e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 497,498...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
6.0021 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 497,498...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.84030e+03+6.05262e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -4.47446e+02+7.04236e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.41652e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +4.47446e+02+7.04236e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.84030e+03-6.05262e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 499,498...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.41304e+04-8.33826e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.65308e+03+7.24424e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.32082e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.65308e+03+7.24424e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.41304e+04+8.33826e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 500,498...
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.00696e+03-6.39027e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.22198e+04-1.04451e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.10264e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.22198e+04-1.04451e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.00696e+03+6.39027e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 501,498...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -9.31229e+02+4.60518e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.54511e+03+3.61175e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.41194e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.54511e+03+3.61175e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -9.31229e+02-4.60518e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 502,498...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.66828e+03+8.89163e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -6.42231e+02+1.60464e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.44891e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +6.42231e+02+1.60464e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.66828e+03-8.89163e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 19,499...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.57405e+03+3.70358e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.90868e+03+4.49443e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.31119e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.90868e+03+4.49443e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.57405e+03-3.70358e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 23,499...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.75638e+02-1.10817e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +8.84980e+02+1.69964e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.81655e+00 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -8.84980e+02+1.69964e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.75638e+02+1.10817e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 25,499...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.85373e+03-2.33203e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.38027e+03-4.94698e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.28713e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.38027e+03-4.94698e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.85373e+03+2.33203e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 27,499...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.35568e+03-3.61071e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.60440e+02+3.75977e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.12694e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.60440e+02+3.75977e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.35568e+03+3.61071e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 28,499...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -7.96008e+02+1.25645e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -6.06038e+02+3.33477e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.08311e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +6.06038e+02+3.33477e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -7.96008e+02-1.25645e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 455,499...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.39415e+02+8.07634e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.30630e+03-9.29338e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.21575e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.30630e+03-9.29338e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.39415e+02-8.07634e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 458,499...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.62837e+02+1.49867e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +7.94879e+02-6.67605e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +9.53223e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -7.94879e+02-6.67605e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.62837e+02-1.49867e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 493,499...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.20898e+03-1.01560e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +9.02756e+02+3.28861e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.05053e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -9.02756e+02+3.28861e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.20898e+03+1.01560e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 495,499...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.87721e+01-6.51525e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.32663e+03+1.28895e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.61075e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.32663e+03+1.28895e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.87721e+01+6.51525e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 497,499...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
4.7846 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 497,499...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.25818e+03-1.84620e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.21035e+02+1.29674e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.42347e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.21035e+02+1.29674e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.25818e+03+1.84620e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 498,499...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 498,499...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.41304e+04-8.33826e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.65308e+03+7.24424e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.32082e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.65308e+03+7.24424e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.41304e+04+8.33826e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 500,499...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.51087e+03+2.31920e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.37703e+04-4.13755e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.66230e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.37703e+04-4.13755e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.51087e+03-2.31920e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 501,499...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.98338e+01+1.09634e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.33961e+03+2.15575e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.85808e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.33961e+03+2.15575e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.98338e+01-1.09634e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 502,499...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.74534e+03+6.24733e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.62809e+03+2.75342e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.99064e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.62809e+03+2.75342e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.74534e+03-6.24733e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 503,499...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +8.04913e+02+1.23514e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.89504e+02+1.45685e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.43392e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +7.89504e+02+1.45685e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +8.04913e+02-1.23514e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 19,500...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +9.64802e+02+5.47565e+00i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.95555e+00+2.80353e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.53802e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +7.95555e+00+2.80353e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +9.64802e+02-5.47565e+00i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 21,500...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.67825e+03-4.07583e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +9.03116e+00+7.43839e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.23612e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -9.03116e+00+7.43839e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.67825e+03+4.07583e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 25,500...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.64044e+03-6.84389e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.11799e+03+1.41763e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.55721e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.11799e+03+1.41763e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.64044e+03+6.84389e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 27,500...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.45624e+02-1.43354e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.16016e+03+9.78286e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.41103e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.16016e+03+9.78286e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.45624e+02+1.43354e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 28,500...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.26065e+02+9.14967e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.40773e+03+8.97350e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.03183e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.40773e+03+8.97350e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.26065e+02-9.14967e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 38,500...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.11517e+03-3.17284e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.50312e+03+2.09670e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.35641e+02 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.50312e+03+2.09670e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.11517e+03+3.17284e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 43,500...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.25820e+02+1.62108e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.55925e+03-2.40654e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.39765e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.55925e+03-2.40654e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.25820e+02-1.62108e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 44,500...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.19049e+02+5.99402e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.24513e+03-1.94889e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -8.06061e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.24513e+03-1.94889e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.19049e+02-5.99402e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 45,500...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.92332e+02+1.47440e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.89676e+02-1.21761e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.68594e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.89676e+02-1.21761e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.92332e+02-1.47440e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 493,500...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.67012e+03-3.02425e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -8.82754e+02-5.20924e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.69665e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +8.82754e+02-5.20924e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.67012e+03+3.02425e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 495,500...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.55124e+03-3.04852e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.53173e+03+6.83608e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -8.14015e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.53173e+03+6.83608e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.55124e+03+3.04852e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 496,500...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.71734e+03+2.43967e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.05800e+02+1.49697e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +8.86222e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.05800e+02+1.49697e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.71734e+03-2.43967e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 497,500...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.0576 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 497,500...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.61114e+03+8.50885e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.83211e+02-2.41281e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.35508e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.83211e+02-2.41281e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.61114e+03-8.50885e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 498,500...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 498,500...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.00696e+03-6.39027e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.22198e+04-1.04451e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.10264e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.22198e+04-1.04451e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.00696e+03+6.39027e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 499,500...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 499,500...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.51087e+03+2.31920e+03i Hz
```

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[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.37703e+04-4.13755e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.66230e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.37703e+04-4.13755e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.51087e+03-2.31920e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 501,500...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.54878e+03+3.75650e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.49186e+03+1.66797e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.41423e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.49186e+03+1.66797e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.54878e+03-3.75650e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 502,500...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.28068e+03+2.47156e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.18321e+03-7.19519e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.99160e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.18321e+03-7.19519e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.28068e+03-2.47156e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 503,500...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.92993e+02+1.55580e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.24526e+02+1.59892e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.29975e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.24526e+02+1.59892e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.92993e+02-1.55580e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 25,501...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.67893e+03+7.31044e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.79056e+02+1.33988e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.07754e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.79056e+02+1.33988e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.67893e+03-7.31044e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 44,501...
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```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -6.43599e+02+1.45908e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.11002e+03-7.23582e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +8.52627e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.11002e+03-7.23582e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -6.43599e+02-1.45908e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 194,501...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +6.10880e+02-1.54556e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.43477e+02-1.09330e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +9.27527e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +7.43477e+02-1.09330e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +6.10880e+02+1.54556e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 323,501...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +6.15179e+02+4.41270e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.68194e+02+1.45598e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -6.43186e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.68194e+02+1.45598e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +6.15179e+02-4.41270e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 326,501...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.67671e+02+1.63776e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.88152e+03+2.65990e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.01264e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.88152e+03+2.65990e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.67671e+02-1.63776e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 330,501...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.98301e+01-2.53234e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +8.02311e+02+1.09416e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.61365e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -8.02311e+02+1.09416e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.98301e+01+2.53234e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 383,501...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.76174e+02+1.58077e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.99168e+02+5.70349e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.25616e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.99168e+02+5.70349e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.76174e+02-1.58077e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 495,501...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.97361e+03-8.49096e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.39747e+03+1.47642e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.11928e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.39747e+03+1.47642e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.97361e+03+8.49096e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 496,501...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.26702e+02-4.67700e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.19943e+03+1.16737e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.57571e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.19943e+03+1.16737e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.26702e+02+4.67700e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 497,501...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.1894 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 497,501...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.72305e+02-1.55504e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.55073e+03-2.01213e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.38928e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.55073e+03-2.01213e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.72305e+02+1.55504e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 498,501...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -9.31229e+02+4.60518e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.54511e+03+3.61175e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.41194e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.54511e+03+3.61175e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -9.31229e+02-4.60518e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 499,501...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.98338e+01+1.09634e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.33961e+03+2.15575e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.85808e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.33961e+03+2.15575e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.98338e+01-1.09634e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 500,501...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.54878e+03+3.75650e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.49186e+03+1.66797e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.41423e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.49186e+03+1.66797e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.54878e+03-3.75650e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 502,501...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.11837e+03+8.79909e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +8.72663e+03-3.02144e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.33049e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -8.72663e+03-3.02144e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.11837e+03-8.79909e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 503,501...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.32262e+04+4.14921e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.78524e+03-1.16549e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +7.22362e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.78524e+03-1.16549e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.32262e+04-4.14921e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 25,502...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.48836e+03+1.15622e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.56347e+02-4.56113e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.48848e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.56347e+02-4.56113e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.48836e+03-1.15622e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 27,502...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.56142e+03+9.81655e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.50641e+01+7.98128e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.11103e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.50641e+01+7.98128e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.56142e+03-9.81655e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 382,502...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.04981e+01+1.37483e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.29994e+02+5.37955e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +9.53344e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.29994e+02+5.37955e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.04981e+01-1.37483e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 384,502...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.27674e+03+8.95317e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.29117e+02+7.25780e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.12157e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.29117e+02+7.25780e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.27674e+03-8.95317e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 424,502...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.25122e+03-1.26053e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.89982e+03+7.91060e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.76701e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.89982e+03+7.91060e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.25122e+03+1.26053e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 425,502...
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -6.94806e+02-1.80683e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.81484e+02+1.24651e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.57037e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.81484e+02+1.24651e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -6.94806e+02+1.80683e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 455,502...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.43912e+01-8.66549e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.81173e+02+7.67342e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.12849e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.81173e+02+7.67342e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.43912e+01+8.66549e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 456,502...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -7.68207e+02+8.03180e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.55793e+03-6.65720e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.46878e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.55793e+03-6.65720e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -7.68207e+02-8.03180e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 457,502...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.48151e+02+1.39914e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +9.73192e+02-1.42649e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.16790e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -9.73192e+02-1.42649e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.48151e+02-1.39914e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 458,502...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.65454e+01+2.09905e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.29135e+03-4.21729e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.32563e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -4.29135e+03-4.21729e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.65454e+01-2.09905e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 495,502...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.06340e+03-7.81167e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.42993e+03+5.24501e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +8.66835e+01 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.42993e+03+5.24501e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.06340e+03+7.81167e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 496,502...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.26737e+02-1.16648e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.52116e+03+1.06327e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.17962e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.52116e+03+1.06327e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.26737e+02+1.16648e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 497,502...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
5.062 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 497,502...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.16227e+03-4.67055e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.92125e+03-1.01938e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.26298e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.92125e+03-1.01938e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.16227e+03+4.67055e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 498,502...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.66828e+03+8.89163e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -6.42231e+02+1.60464e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.44891e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +6.42231e+02+1.60464e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.66828e+03-8.89163e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 499,502...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.74534e+03+6.24733e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.62809e+03+2.75342e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.99064e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +3.62809e+03+2.75342e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.74534e+03-6.24733e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 500,502...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.28068e+03+2.47156e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.18321e+03-7.19519e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.99160e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.18321e+03-7.19519e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.28068e+03-2.47156e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 501,502...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 501,502...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.11837e+03+8.79909e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +8.72663e+03-3.02144e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.33049e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -8.72663e+03-3.02144e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.11837e+03-8.79909e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 503,502...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.05562e+04-3.71004e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.57067e+03+1.50673e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +6.11628e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.57067e+03+1.50673e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.05562e+04+3.71004e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 323,503...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +9.08049e+01+1.65523e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -5.72047e+02+9.66296e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.57255e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +5.72047e+02+9.66296e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +9.08049e+01-1.65523e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 344,503...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.54450e+03-1.14658e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.26715e+02-2.45097e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.35945e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.26715e+02-2.45097e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.54450e+03+1.14658e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 382,503...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.92097e+03+2.06612e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.66439e+01+3.33933e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.30246e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +7.66439e+01+3.33933e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.92097e+03-2.06612e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 383,503...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -9.14615e+02+3.09768e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.25076e+03+2.42968e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.55320e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.25076e+03+2.42968e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -9.14615e+02-3.09768e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 384,503...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.37634e+03+5.38809e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.49040e+02-3.82021e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.79381e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.49040e+02-3.82021e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.37634e+03-5.38809e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 457,503...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.11615e+02+6.73905e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.67598e+03-9.40206e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.26442e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.67598e+03-9.40206e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.11615e+02-6.73905e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 458,503...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.20409e+02-2.85097e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.81171e+03+6.89333e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.22649e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.81171e+03+6.89333e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.20409e+02+2.85097e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 494,503...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.24332e+03-8.85278e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.00618e+02-9.60897e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.21957e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.00618e+02-9.60897e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.24332e+03+8.85278e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 495,503...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.30116e+03+1.43659e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.26982e+03-5.63131e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.17615e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.26982e+03-5.63131e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.30116e+03-1.43659e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 496,503...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.78883e+03-9.82833e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.29358e+03+5.37610e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.62732e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.29358e+03+5.37610e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.78883e+03+9.82833e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 497,503...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
5.5876 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 497,503...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.81966e+03+1.18960e+03i Hz
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```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.99296e+03+9.11631e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.83365e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.99296e+03+9.11631e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.81966e+03-1.18960e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 499,503...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +8.04913e+02+1.23514e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.89504e+02+1.45685e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.43392e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +7.89504e+02+1.45685e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +8.04913e+02-1.23514e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 500,503...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.92993e+02+1.55580e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.24526e+02+1.59892e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.29975e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.24526e+02+1.59892e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.92993e+02-1.55580e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 501,503...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 501,503...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.32262e+04+4.14921e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.78524e+03-1.16549e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +7.22362e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.78524e+03-1.16549e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.32262e+04-4.14921e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 502,503...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 502,503...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.05562e+04-3.71004e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.57067e+03+1.50673e+04i Hz
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```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +6.11628e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.57067e+03+1.50673e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.05562e+04+3.71004e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 331,504...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.09149e+03+1.51644e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.97033e+03+1.52179e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -9.08255e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.97033e+03+1.52179e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.09149e+03-1.51644e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 333,504...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.10380e+03+5.53621e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -3.37177e+02+1.42434e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.99966e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +3.37177e+02+1.42434e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.10380e+03-5.53621e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 343,504...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.93776e+02+6.26688e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +8.92359e+02-1.40386e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.06705e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -8.92359e+02-1.40386e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.93776e+02-6.26688e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 494,504...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.14161e+03-3.14351e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +7.07373e+02+5.23346e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -8.64879e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -7.07373e+02+5.23346e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.14161e+03+3.14351e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 495,504...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.69664e+03-8.71692e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.43677e+02+5.94049e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.47749e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.43677e+02+5.94049e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.69664e+03+8.71692e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 496,504...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.88140e+02-4.78604e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.44058e+03+1.62844e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.53697e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.44058e+03+1.62844e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.88140e+02+4.78604e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 505,504...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.97737e+03+1.62537e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.87187e+03-1.02884e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +6.05678e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.87187e+03-1.02884e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.97737e+03-1.62537e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 506,504...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.19681e+03-4.24143e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.55030e+03+4.19298e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.84154e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.55030e+03+4.19298e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.19681e+03+4.24143e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 507,504...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.62750e+02-1.00854e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.49295e+03+9.57591e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.51396e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.49295e+03+9.57591e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.62750e+02+1.00854e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 42,505...
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.13746e+02-1.35994e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.21946e+02-3.67142e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.20510e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.21946e+02-3.67142e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.13746e+02+1.35994e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 46,505...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.22754e+03+2.40187e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.57511e+02+1.68300e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.96595e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.57511e+02+1.68300e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.22754e+03-2.40187e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 48,505...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.67940e+03+2.01856e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.33428e+03+6.25593e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.80634e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.33428e+03+6.25593e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.67940e+03-2.01856e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 504,505...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.7698 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 504,505...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.97737e+03+1.62537e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.87187e+03-1.02884e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +6.05678e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.87187e+03-1.02884e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.97737e+03-1.62537e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 506,505...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.07002e+03+2.42047e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.46794e+02-7.52905e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.51239e+03 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.46794e+02-7.52905e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.07002e+03-2.42047e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 507,505...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.04115e+03-8.37807e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.70451e+01+3.25488e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.25876e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.70451e+01+3.25488e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.04115e+03+8.37807e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 508,505...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -7.59001e+02-2.47825e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -5.06768e+02-3.74797e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.05368e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +5.06768e+02-3.74797e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -7.59001e+02+2.47825e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 510,505...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.16145e+02-4.63787e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.99618e+03-3.77655e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.28135e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.99618e+03-3.77655e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.16145e+02+4.63787e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 331,506...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.91880e+03-1.13184e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.15307e+03-3.14742e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.55714e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.15307e+03-3.14742e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.91880e+03+1.13184e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 333,506...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.03339e+03+5.25301e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -3.25825e+03+2.23260e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.46850e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +3.25825e+03+2.23260e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.03339e+03-5.25301e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 336,506...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +9.74540e+02+1.08377e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.09706e+02-2.46187e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.16969e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.09706e+02-2.46187e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +9.74540e+02-1.08377e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 338,506...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.28330e+03+2.70921e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.69801e+03-2.68318e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.07458e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.69801e+03-2.68318e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.28330e+03-2.70921e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 504,506...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.19681e+03-4.24143e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.55030e+03+4.19298e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.84154e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.55030e+03+4.19298e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.19681e+03+4.24143e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 505,506...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
12.6883 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 505,506...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.07002e+03+2.42047e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.46794e+02-7.52905e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.51239e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.46794e+02-7.52905e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.07002e+03-2.42047e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 507,506...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.13963e+03+4.38359e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.59007e+04-4.00942e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.32622e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.59007e+04-4.00942e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.13963e+03-4.38359e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 508,506...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.35391e+03+5.97859e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.36279e-01+1.55328e-01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.20844e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +7.36279e-01+1.55328e-01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.35391e+03-5.97859e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 333,507...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.14823e+03+1.58519e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.59244e+02-1.83369e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.56425e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -3.59244e+02-1.83369e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.14823e+03-1.58519e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 504,507...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.62750e+02-1.00854e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.49295e+03+9.57591e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.51396e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.49295e+03+9.57591e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.62750e+02+1.00854e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 505,507...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
2.4745 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 505,507...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.04115e+03-8.37807e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.70451e+01+3.25488e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.25876e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.70451e+01+3.25488e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.04115e+03+8.37807e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 506,507...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 506,507...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.13963e+03+4.38359e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.59007e+04-4.00942e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.32622e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.59007e+04-4.00942e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.13963e+03-4.38359e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 508,507...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.96309e+03+1.17153e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.79495e+03+5.32474e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.56436e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.79495e+03+5.32474e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.96309e+03-1.17153e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 510,507...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.22991e+03-4.33530e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.99331e+02-2.06976e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.07779e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.99331e+02-2.06976e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.22991e+03+4.33530e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 529,507...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.97123e+00+2.80735e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -8.30155e+02+8.48000e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.81817e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +8.30155e+02+8.48000e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.97123e+00-2.80735e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 48,508...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.55703e+03+2.71430e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.92103e+01+7.40845e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.01216e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.92103e+01+7.40845e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.55703e+03-2.71430e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 49,508...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.21810e+03+1.33982e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.20563e+02-7.24684e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.33692e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -3.20563e+02-7.24684e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.21810e+03-1.33982e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 505,508...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -7.59001e+02-2.47825e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.06768e+02-3.74797e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.05368e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.06768e+02-3.74797e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -7.59001e+02+2.47825e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 506,508...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.35391e+03+5.97859e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.36279e-01+1.55328e-01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.20844e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +7.36279e-01+1.55328e-01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.35391e+03-5.97859e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 507,508...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.96309e+03+1.17153e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.79495e+03+5.32474e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.56436e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.79495e+03+5.32474e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.96309e+03-1.17153e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 490,509...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.62629e+03+7.68733e+00i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.74548e-01-1.16164e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.32449e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.74548e-01-1.16164e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.62629e+03-7.68733e+00i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 40,510...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.80840e+03-5.83172e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +7.89591e+01+5.02117e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.49592e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -7.89591e+01+5.02117e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.80840e+03+5.83172e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 42,510...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +6.84129e+02-1.36760e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +7.74251e+02+1.25303e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +6.69344e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -7.74251e+02+1.25303e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +6.84129e+02+1.36760e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 46,510...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.68108e+02+6.98478e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -9.07819e+02+1.70124e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.11884e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +9.07819e+02+1.70124e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.68108e+02-6.98478e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 57,510...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.80543e+02+1.10145e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.68449e+03-1.10291e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.96003e+02 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.68449e+03-1.10291e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.80543e+02-1.10145e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 505,510...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.16145e+02-4.63787e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.99618e+03-3.77655e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.28135e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.99618e+03-3.77655e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.16145e+02+4.63787e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 507,510...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.22991e+03-4.33530e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.99331e+02-2.06976e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.07779e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.99331e+02-2.06976e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.22991e+03+4.33530e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 511,510...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.85892e+03+5.86923e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.50563e+03+2.54543e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.49572e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.50563e+03+2.54543e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.85892e+03-5.86923e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 512,510...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.07424e+02-3.21962e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.58851e+03-1.80220e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.92376e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.58851e+03-1.80220e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.07424e+02+3.21962e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 513,510...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.03019e+03-1.10510e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.38495e+03-5.44113e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.68041e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.38495e+03-5.44113e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.03019e+03+1.10510e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 321,511...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.30137e+03+2.10766e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.09497e+03-1.20308e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.10555e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.09497e+03-1.20308e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.30137e+03-2.10766e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 328,511...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.64455e+02-1.28152e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +5.79000e+02+6.58050e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +8.12183e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -5.79000e+02+6.58050e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.64455e+02+1.28152e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 329,511...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.17642e+03-5.44829e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.70221e+03+2.50775e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.39895e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.70221e+03+2.50775e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.17642e+03+5.44829e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 330,511...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.92704e+02-8.93802e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.40320e+03+1.01700e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.35704e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.40320e+03+1.01700e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.92704e+02+8.93802e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 331,511...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.78445e+02+6.96541e+02i Hz
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```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +9.34385e+02-1.37980e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -8.98815e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -9.34385e+02-1.37980e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.78445e+02-6.96541e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 510,511...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.4684 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 510,511...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.85892e+03+5.86923e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.50563e+03+2.54543e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.49572e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.50563e+03+2.54543e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.85892e+03-5.86923e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 512,511...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.24486e+03+3.93297e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.83595e+02+7.23354e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.68232e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.83595e+02+7.23354e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.24486e+03-3.93297e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 513,511...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -7.09708e+02+2.92035e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.22176e+03-9.60409e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.12579e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.22176e+03-9.60409e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -7.09708e+02-2.92035e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 514,511...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.62936e+02+6.44697e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.44423e+03-2.47040e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.91451e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.44423e+03-2.47040e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.62936e+02-6.44697e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 515,511...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.38152e+02-8.01269e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -8.09971e+02-1.36607e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.81848e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +8.09971e+02-1.36607e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.38152e+02+8.01269e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 516,511...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.01607e+03-2.12936e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -6.28128e+02-1.45971e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.04268e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +6.28128e+02-1.45971e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.01607e+03+2.12936e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 517,511...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.12909e+03-4.33468e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.76763e+03-4.45773e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.18559e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.76763e+03-4.45773e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.12909e+03+4.33468e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 521,511...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.80589e+03+1.32443e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.82724e+03+8.15180e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -6.64935e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.82724e+03+8.15180e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.80589e+03-1.32443e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 40,512...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +6.74643e+02+3.85201e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.85721e+03+4.59148e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.60955e+02 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +3.85721e+03+4.59148e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +6.74643e+02-3.85201e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 41,512...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.57842e+03-1.72444e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.25342e+03+5.48826e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.38171e+01 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.25342e+03+5.48826e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.57842e+03+1.72444e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 42,512...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.48777e+02+7.61977e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.65509e+02+2.45928e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -6.70532e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.65509e+02+2.45928e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.48777e+02-7.61977e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 233,512...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +8.20978e+02-1.59535e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.48244e+01-4.06934e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.46444e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.48244e+01-4.06934e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +8.20978e+02+1.59535e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 328,512...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.66783e+03-5.09078e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.11474e+03+1.66340e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.12640e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.11474e+03+1.66340e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.66783e+03+5.09078e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 329,512...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.27525e+03+2.09321e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.32706e+03-5.17584e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.25639e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.32706e+03-5.17584e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.27525e+03-2.09321e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 330,512...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.36423e+02+3.98457e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.32951e+03-3.00499e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -6.21940e+01 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.32951e+03-3.00499e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.36423e+02-3.98457e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 510,512...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.07424e+02-3.21962e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.58851e+03-1.80220e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.92376e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.58851e+03-1.80220e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.07424e+02+3.21962e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 511,512...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
3.1025 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 511,512...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.24486e+03+3.93297e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.83595e+02+7.23354e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.68232e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.83595e+02+7.23354e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.24486e+03-3.93297e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 513,512...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -9.32251e+03-7.53185e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.40635e+04+4.97124e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.20666e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.40635e+04+4.97124e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -9.32251e+03+7.53185e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 514,512...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.0203 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 514,512...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.51012e+02+1.83315e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -4.45972e+02+2.28162e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.13467e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +4.45972e+02+2.28162e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.51012e+02-1.83315e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 515,512...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.53324e+02-1.22425e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.92999e+03-1.37741e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.06910e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.92999e+03-1.37741e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.53324e+02+1.22425e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 516,512...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.52222e+03-2.18743e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -5.34152e+03-1.01939e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.81724e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +5.34152e+03-1.01939e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.52222e+03+2.18743e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 517,512...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.25589e+03+1.84855e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.43460e+03+1.28928e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.38288e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.43460e+03+1.28928e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.25589e+03-1.84855e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 518,512...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.32145e+02-9.39677e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.97564e+02-9.31933e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.27719e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.97564e+02-9.31933e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.32145e+02+9.39677e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 519,512...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.54783e+02+6.25787e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.89030e+03-1.47971e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.12317e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.89030e+03-1.47971e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.54783e+02-6.25787e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 520,512...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.11658e+03-3.61960e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -4.04773e+02-4.76822e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.60046e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +4.04773e+02-4.76822e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.11658e+03+3.61960e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 40,513...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.17774e+03-1.92905e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.35112e+02+2.88998e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.90151e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.35112e+02+2.88998e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.17774e+03+1.92905e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 41,513...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.68594e+02-9.34865e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +8.94631e+02+1.18786e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.33948e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -8.94631e+02+1.18786e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.68594e+02+9.34865e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 42,513...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.11106e+01-3.07181e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +8.70924e+02+8.13125e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.63094e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -8.70924e+02+8.13125e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.11106e+01+3.07181e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 57,513...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.60590e+03-2.11518e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.15468e+03+2.64995e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.64655e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.15468e+03+2.64995e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.60590e+03+2.11518e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 59,513...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.27478e+02+2.20893e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +8.25775e+02-1.00090e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.53153e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -8.25775e+02-1.00090e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.27478e+02-2.20893e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 510,513...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.03019e+03-1.10510e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.38495e+03-5.44113e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.68041e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.38495e+03-5.44113e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.03019e+03+1.10510e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 511,513...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
12.8975 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 511,513...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -7.09708e+02+2.92035e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.22176e+03-9.60409e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.12579e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.22176e+03-9.60409e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -7.09708e+02-2.92035e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 512,513...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 512,513...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -9.32251e+03-7.53185e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.40635e+04+4.97124e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.20666e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.40635e+04+4.97124e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -9.32251e+03+7.53185e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 514,513...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
4.6984 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 514,513...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.84302e+03+1.74413e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -4.99857e+03+1.99023e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.58539e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +4.99857e+03+1.99023e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.84302e+03-1.74413e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 516,513...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.70088e+03-9.34148e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -4.07970e+01-1.04659e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.58405e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +4.07970e+01-1.04659e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.70088e+03+9.34148e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 517,513...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.40890e+03+5.74603e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -5.41543e+02+1.06185e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.16065e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +5.41543e+02+1.06185e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.40890e+03-5.74603e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 518,513...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -6.69206e+01-8.55831e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.40216e+03-1.29680e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.03384e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.40216e+03-1.29680e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -6.69206e+01+8.55831e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 519,513...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.50743e+03-8.44307e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.10189e+03-6.72534e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.00625e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.10189e+03-6.72534e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.50743e+03+8.44307e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 520,513...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.10878e+03-2.88845e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.33705e+03-1.94543e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.79091e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.33705e+03-1.94543e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.10878e+03+2.88845e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 308,514...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.79195e+02-1.02688e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.44235e+03-8.42430e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.51464e+00 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.44235e+03-8.42430e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.79195e+02+1.02688e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 312,514...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -6.38796e+02+1.37109e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.03790e+02+3.21068e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.13871e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.03790e+02+3.21068e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -6.38796e+02-1.37109e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 511,514...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.62936e+02+6.44697e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.44423e+03-2.47040e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.91451e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.44423e+03-2.47040e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.62936e+02-6.44697e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 512,514...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.51012e+02+1.83315e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.45972e+02+2.28162e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.13467e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.45972e+02+2.28162e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.51012e+02-1.83315e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 513,514...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.84302e+03+1.74413e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.99857e+03+1.99023e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.58539e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.99857e+03+1.99023e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.84302e+03-1.74413e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 515,514...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.54266e+03-4.93723e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.05183e+02+2.47090e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.51786e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -4.05183e+02+2.47090e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.54266e+03+4.93723e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 516,514...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.76039e+02-2.04744e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.41789e+03+2.01426e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.42514e+02 Hz
```

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[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.41789e+03+2.01426e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.76039e+02+2.04744e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 517,514...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.94720e+03+3.83492e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +5.54880e+03-2.14741e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.51454e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -5.54880e+03-2.14741e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.94720e+03-3.83492e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 518,514...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.49789e+03-6.75943e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.43056e+02-3.96875e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.35760e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.43056e+02-3.96875e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.49789e+03+6.75943e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 519,514...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.15772e+03+5.39354e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -3.13426e+01+4.62964e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.74291e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +3.13426e+01+4.62964e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.15772e+03-5.39354e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 520,514...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.01634e+03+1.53349e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -4.27461e+01+1.68270e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.08303e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +4.27461e+01+1.68270e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.01634e+03-1.53349e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 521,514...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.79482e+03+2.63619e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +9.44354e+02-2.37747e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.44154e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -9.44354e+02-2.37747e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.79482e+03-2.63619e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 522,514...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.17813e+03-1.07200e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.67291e+02+4.32425e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.24546e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.67291e+02+4.32425e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.17813e+03+1.07200e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 534,514...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.24161e+02-5.90109e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.33323e+03+9.19730e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.18121e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.33323e+03+9.19730e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.24161e+02+5.90109e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 538,514...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +9.97675e+01+3.71882e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.09059e+03-1.42174e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.08994e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.09059e+03-1.42174e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +9.97675e+01-3.71882e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 539,514...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.89630e+01+9.81866e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -7.55836e+02+1.22152e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -7.58807e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +7.55836e+02+1.22152e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.89630e+01-9.81866e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 231,515...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.85474e+03+3.11019e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.60186e+01+5.52690e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.46877e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.60186e+01+5.52690e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.85474e+03-3.11019e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 234,515...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +6.85260e+02-7.97595e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +6.68988e+02+1.45675e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.39067e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -6.68988e+02+1.45675e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +6.85260e+02+7.97595e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 305,515...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.73222e+02+9.46482e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.18132e+03+1.91139e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.08386e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.18132e+03+1.91139e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.73222e+02-9.46482e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 306,515...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.24634e+03+5.32304e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.28701e+03+2.74713e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.76238e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.28701e+03+2.74713e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.24634e+03-5.32304e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 307,515...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.98128e+03+3.09565e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.56380e+03+8.67580e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.68351e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.56380e+03+8.67580e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.98128e+03-3.09565e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 308,515...
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.23823e+02-2.62381e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.82468e+03-1.30322e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.11737e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.82468e+03-1.30322e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.23823e+02+2.62381e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 309,515...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.75500e+02-2.85783e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.27432e+03-5.41570e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.64764e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.27432e+03-5.41570e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.75500e+02+2.85783e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 312,515...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.49738e+03+1.01967e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.60321e+02+8.44782e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.30306e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.60321e+02+8.44782e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.49738e+03-1.01967e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 328,515...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.64955e+01-6.47913e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +7.30444e+02+2.43240e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.15079e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -7.30444e+02+2.43240e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.64955e+01+6.47913e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 329,515...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.09904e+02+1.34725e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.64659e+02-1.24325e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.54393e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -3.64659e+02-1.24325e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.09904e+02-1.34725e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 511,515...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.38152e+02-8.01269e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -8.09971e+02-1.36607e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.81848e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +8.09971e+02-1.36607e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.38152e+02+8.01269e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 512,515...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.53324e+02-1.22425e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.92999e+03-1.37741e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.06910e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.92999e+03-1.37741e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.53324e+02+1.22425e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 514,515...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
4.7694 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 514,515...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.54266e+03-4.93723e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.05183e+02+2.47090e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.51786e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -4.05183e+02+2.47090e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.54266e+03+4.93723e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 516,515...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.33680e+02+7.51079e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.69292e+03+4.19088e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.52941e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.69292e+03+4.19088e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.33680e+02-7.51079e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 517,515...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.04703e+04-9.76448e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.43694e+02-1.59837e+04i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.35369e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +7.43694e+02-1.59837e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.04703e+04+9.76448e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 518,515...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.36762e+03+4.21021e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +4.03659e+03-1.94029e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.88312e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -4.03659e+03-1.94029e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.36762e+03-4.21021e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 519,515...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.71473e+03+7.55883e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +9.68822e+01-2.04063e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.52793e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -9.68822e+01-2.04063e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.71473e+03-7.55883e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 520,515...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.19961e+03+3.51730e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.55052e+03+1.10940e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.63499e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.55052e+03+1.10940e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.19961e+03-3.51730e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 539,515...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.28261e+02-7.91828e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.30601e+03+1.11149e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -8.41890e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.30601e+03+1.11149e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.28261e+02+7.91828e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 41,516...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.15852e+01+1.18630e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.81043e+03+1.74808e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.20902e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.81043e+03+1.74808e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.15852e+01-1.18630e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 231,516...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.81042e+03-1.19644e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.26608e+00-1.29107e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.10236e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.26608e+00-1.29107e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.81042e+03+1.19644e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 233,516...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.28377e+03+3.68534e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.22889e+02+2.19685e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.09989e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.22889e+02+2.19685e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.28377e+03-3.68534e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 234,516...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.39858e+03-2.84759e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +6.59407e+02+1.05851e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.39021e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -6.59407e+02+1.05851e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.39858e+03+2.84759e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 235,516...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +8.55518e+02-7.18678e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.73145e+02+1.57334e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.12200e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.73145e+02+1.57334e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +8.55518e+02+7.18678e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 307,516...
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -9.84674e+02+6.60781e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.03223e+03+6.18683e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.85361e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.03223e+03+6.18683e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -9.84674e+02-6.60781e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 324,516...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -9.87838e+02+1.58151e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +9.94233e+02-5.51231e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.23956e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -9.94233e+02-5.51231e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -9.87838e+02-1.58151e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 328,516...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.72120e+03-2.89763e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.46599e+03+4.56883e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.61183e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.46599e+03+4.56883e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.72120e+03+2.89763e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 329,516...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.35790e+03+2.72764e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.06047e+03-6.69106e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.18595e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -3.06047e+03-6.69106e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.35790e+03-2.72764e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 330,516...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.48092e+01+3.19115e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +9.62264e+02-1.00796e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.22082e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -9.62264e+02-1.00796e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.48092e+01-3.19115e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 511,516...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.01607e+03-2.12936e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -6.28128e+02-1.45971e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.04268e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +6.28128e+02-1.45971e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.01607e+03+2.12936e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 512,516...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.52222e+03-2.18743e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.34152e+03-1.01939e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.81724e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.34152e+03-1.01939e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.52222e+03+2.18743e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 513,516...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.70088e+03-9.34148e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.07970e+01-1.04659e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.58405e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.07970e+01-1.04659e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.70088e+03+9.34148e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 514,516...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.0606 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 514,516...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.76039e+02-2.04744e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.41789e+03+2.01426e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.42514e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.41789e+03+2.01426e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.76039e+02+2.04744e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 515,516...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 515,516...
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.33680e+02+7.51079e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.69292e+03+4.19088e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.52941e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.69292e+03+4.19088e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.33680e+02-7.51079e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 517,516...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.35610e+04+4.13602e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.65390e+03+1.10920e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +7.95450e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.65390e+03+1.10920e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.35610e+04-4.13602e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 518,516...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -6.03464e+02+6.47182e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.29906e+03-9.99709e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.17719e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.29906e+03-9.99709e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -6.03464e+02-6.47182e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 519,516...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.24043e+03+4.09974e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.41806e+03-2.28268e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.22024e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.41806e+03-2.28268e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.24043e+03-4.09974e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 520,516...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.46813e+03+4.24391e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.12334e+03-2.22447e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.32992e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -3.12334e+03-2.22447e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.46813e+03-4.24391e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 305,517...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.44975e+02+1.69576e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.00781e+03+4.64309e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.07092e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.00781e+03+4.64309e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.44975e+02-1.69576e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 307,517...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.78768e+03-1.63221e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.14596e+03-1.22015e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.63276e+01 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.14596e+03-1.22015e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.78768e+03+1.63221e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 308,517...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.92681e+02-2.10725e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.34833e+02-1.34815e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.97697e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.34833e+02-1.34815e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.92681e+02+2.10725e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 312,517...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.05311e+03+1.73007e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.70083e+03+3.02645e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -7.75587e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.70083e+03+3.02645e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.05311e+03-1.73007e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 321,517...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.10463e+02+5.93356e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +8.26588e+02-1.80149e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.41000e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -8.26588e+02-1.80149e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.10463e+02-5.93356e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 324,517...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.00028e+03+2.50413e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.47467e+03-1.54894e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.83518e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.47467e+03-1.54894e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.00028e+03-2.50413e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 328,517...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.65444e+02-7.35404e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.15594e+03+1.86522e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.72332e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.15594e+03+1.86522e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.65444e+02+7.35404e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 329,517...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.05096e+02-1.71890e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.52876e+03+9.19550e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.08271e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.52876e+03+9.19550e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.05096e+02+1.71890e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 511,517...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.12909e+03-4.33468e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.76763e+03-4.45773e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.18559e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.76763e+03-4.45773e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.12909e+03+4.33468e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 512,517...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.25589e+03+1.84855e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.43460e+03+1.28928e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.38288e+02 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.43460e+03+1.28928e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.25589e+03-1.84855e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 513,517...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.40890e+03+5.74603e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -5.41543e+02+1.06185e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.16065e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +5.41543e+02+1.06185e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.40890e+03-5.74603e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 514,517...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
6.0228 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 514,517...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.94720e+03+3.83492e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +5.54880e+03-2.14741e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.51454e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -5.54880e+03-2.14741e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.94720e+03-3.83492e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 515,517...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 515,517...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.04703e+04-9.76448e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -7.43694e+02-1.59837e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.35369e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +7.43694e+02-1.59837e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.04703e+04+9.76448e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 516,517...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 516,517...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.35610e+04+4.13602e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.65390e+03+1.10920e+04i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +7.95450e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.65390e+03+1.10920e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.35610e+04-4.13602e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 518,517...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.45455e+02+1.26888e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.06892e+03-1.29551e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.64145e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.06892e+03-1.29551e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.45455e+02-1.26888e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 520,517...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.44558e+02+1.65556e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.99066e+02-2.75036e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.39601e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.99066e+02-2.75036e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.44558e+02-1.65556e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 521,517...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.08045e+03-3.81075e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +4.83622e+02+1.01281e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.89597e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -4.83622e+02+1.01281e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.08045e+03+3.81075e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 231,518...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.04950e+02-1.68848e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +9.51036e+02+1.42766e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +8.37403e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -9.51036e+02+1.42766e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.04950e+02+1.68848e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 274,518...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.06283e+02-2.16987e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.66054e+03-4.15647e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.22106e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.66054e+03-4.15647e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.06283e+02+2.16987e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 275,518...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.12045e+03-2.80315e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.28568e+02-1.95358e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.01480e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.28568e+02-1.95358e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.12045e+03+2.80315e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 276,518...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.15562e+02+1.00946e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +7.62101e+02-1.47437e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.01459e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -7.62101e+02-1.47437e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.15562e+02-1.00946e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 277,518...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.96394e+03+7.93943e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.13906e+03-5.85681e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.13115e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.13906e+03-5.85681e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.96394e+03-7.93943e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 308,518...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.41500e+03+1.64337e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.85053e+03+8.77885e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.14176e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.85053e+03+8.77885e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.41500e+03-1.64337e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 512,518...
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.32145e+02-9.39677e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.97564e+02-9.31933e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.27719e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.97564e+02-9.31933e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.32145e+02+9.39677e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 513,518...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -6.69206e+01-8.55831e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.40216e+03-1.29680e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.03384e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.40216e+03-1.29680e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -6.69206e+01+8.55831e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 514,518...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
5.5812 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 514,518...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.49789e+03-6.75943e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.43056e+02-3.96875e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.35760e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.43056e+02-3.96875e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.49789e+03+6.75943e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 515,518...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.36762e+03+4.21021e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.03659e+03-1.94029e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.88312e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -4.03659e+03-1.94029e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.36762e+03-4.21021e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 516,518...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -6.03464e+02+6.47182e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.29906e+03-9.99709e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.17719e+03 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.29906e+03-9.99709e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -6.03464e+02-6.47182e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 517,518...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.45455e+02+1.26888e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.06892e+03-1.29551e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.64145e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.06892e+03-1.29551e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.45455e+02-1.26888e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 519,518...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.22373e+04-3.59096e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.35514e+04-1.94725e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.41302e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.35514e+04-1.94725e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.22373e+04+3.59096e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 520,518...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.83617e+03+1.41866e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -3.69712e+03+1.08322e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.11546e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +3.69712e+03+1.08322e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.83617e+03-1.41866e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 538,518...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.11738e+02-1.42116e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.53575e+03+3.37469e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.70836e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.53575e+03+3.37469e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.11738e+02+1.42116e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 539,518...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.88960e+03+1.02215e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -7.01537e+01+2.59569e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +9.05097e+01 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +7.01537e+01+2.59569e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.88960e+03-1.02215e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 540,518...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.55219e+02+5.00567e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -8.79195e+01+1.25398e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.50542e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +8.79195e+01+1.25398e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.55219e+02-5.00567e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 57,519...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -9.43051e+02-1.30472e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.33829e+03-6.83964e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +7.41623e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.33829e+03-6.83964e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -9.43051e+02+1.30472e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 58,519...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.55815e+03-2.23794e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.66422e+02+1.90351e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.26678e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.66422e+02+1.90351e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.55815e+03+2.23794e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 59,519...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.69718e+03+3.19058e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.80274e+03+1.30070e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.47830e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.80274e+03+1.30070e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.69718e+03-3.19058e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 277,519...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.67961e+02-7.22710e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.27050e+03-1.59964e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.69027e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.27050e+03-1.59964e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.67961e+02+7.22710e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 512,519...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.54783e+02+6.25787e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.89030e+03-1.47971e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.12317e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.89030e+03-1.47971e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.54783e+02-6.25787e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 513,519...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.50743e+03-8.44307e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.10189e+03-6.72534e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.00625e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.10189e+03-6.72534e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.50743e+03+8.44307e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 514,519...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
5.0577 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 514,519...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.15772e+03+5.39354e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.13426e+01+4.62964e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.74291e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.13426e+01+4.62964e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.15772e+03-5.39354e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 515,519...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.71473e+03+7.55883e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +9.68822e+01-2.04063e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.52793e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -9.68822e+01-2.04063e+01i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.71473e+03-7.55883e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 516,519...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.24043e+03+4.09974e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.41806e+03-2.28268e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.22024e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.41806e+03-2.28268e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.24043e+03-4.09974e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 518,519...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 518,519...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.22373e+04-3.59096e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.35514e+04-1.94725e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.41302e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.35514e+04-1.94725e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.22373e+04+3.59096e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 520,519...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.45152e+03-1.18404e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.11701e+04+7.15429e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.13228e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.11701e+04+7.15429e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.45152e+03+1.18404e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 538,519...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.63598e+02+2.91779e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -8.69933e+02+1.48511e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.34218e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +8.69933e+02+1.48511e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.63598e+02-2.91779e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 539,519...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.13228e+02+7.57877e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -8.30641e+02+1.24209e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.41960e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +8.30641e+02+1.24209e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.13228e+02-7.57877e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 40,520...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.29129e+02-9.77003e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.18612e+03-2.42325e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.19868e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.18612e+03-2.42325e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.29129e+02+9.77003e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 41,520...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -9.53472e+00-2.27841e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.89849e+02+2.59465e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.60474e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -3.89849e+02+2.59465e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -9.53472e+00+2.27841e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 59,520...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.91449e+03+2.19177e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.33645e+02+7.62514e+00i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.56959e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.33645e+02+7.62514e+00i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.91449e+03-2.19177e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 231,520...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.82536e+01-4.27030e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.88209e+02-3.92621e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.45777e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.88209e+02-3.92621e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.82536e+01+4.27030e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 232,520...
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.25629e+03-1.01812e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.12430e+01-2.01061e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.30882e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +7.12430e+01-2.01061e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.25629e+03+1.01812e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 233,520...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.86552e+02-1.94183e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.70761e+03+2.11288e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -7.70975e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.70761e+03+2.11288e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.86552e+02+1.94183e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 234,520...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -9.58029e+02-6.99606e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +9.77482e+02+3.18915e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +6.04779e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -9.77482e+02+3.18915e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -9.58029e+02+6.99606e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 275,520...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.86871e+02-2.12004e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.71630e+02-1.55141e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.11351e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.71630e+02-1.55141e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.86871e+02+2.12004e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 277,520...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.40221e+01+1.07003e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.14498e+01-7.15534e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.74806e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.14498e+01-7.15534e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.40221e+01-1.07003e+01i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 512,520...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.11658e+03-3.61960e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.04773e+02-4.76822e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.60046e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.04773e+02-4.76822e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.11658e+03+3.61960e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 513,520...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.10878e+03-2.88845e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.33705e+03-1.94543e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.79091e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.33705e+03-1.94543e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.10878e+03+2.88845e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 514,520...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.1893 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 514,520...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.01634e+03+1.53349e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.27461e+01+1.68270e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.08303e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.27461e+01+1.68270e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.01634e+03-1.53349e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 515,520...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.19961e+03+3.51730e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.55052e+03+1.10940e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.63499e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.55052e+03+1.10940e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.19961e+03-3.51730e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 516,520...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.46813e+03+4.24391e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.12334e+03-2.22447e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.32992e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.12334e+03-2.22447e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.46813e+03-4.24391e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 517,520...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.44558e+02+1.65556e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.99066e+02-2.75036e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.39601e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.99066e+02-2.75036e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.44558e+02-1.65556e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 518,520...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 518,520...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.83617e+03+1.41866e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -3.69712e+03+1.08322e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.11546e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +3.69712e+03+1.08322e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.83617e+03-1.41866e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 519,520...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 519,520...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.45152e+03-1.18404e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.11701e+04+7.15429e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.13228e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.11701e+04+7.15429e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.45152e+03+1.18404e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 312,521...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.59323e+03-9.75232e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -3.91059e+03-7.11015e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -6.57731e+01 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +3.91059e+03-7.11015e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.59323e+03+9.75232e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 314,521...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +6.80169e+01+2.11833e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.25121e+03+1.29203e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.10739e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.25121e+03+1.29203e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +6.80169e+01-2.11833e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 321,521...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.18233e+02-1.91120e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.61402e+03+2.88132e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.53578e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.61402e+03+2.88132e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.18233e+02+1.91120e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 329,521...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.73783e+02-7.49182e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +7.73638e+02+1.25056e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.70802e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -7.73638e+02+1.25056e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.73783e+02+7.49182e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 511,521...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.80589e+03+1.32443e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.82724e+03+8.15180e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -6.64935e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.82724e+03+8.15180e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.80589e+03-1.32443e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 514,521...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.79482e+03+2.63619e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +9.44354e+02-2.37747e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.44154e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -9.44354e+02-2.37747e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.79482e+03-2.63619e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 517,521...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.08045e+03-3.81075e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +4.83622e+02+1.01281e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.89597e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -4.83622e+02+1.01281e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.08045e+03+3.81075e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 522,521...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.76006e+03-2.08250e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.73942e+03+1.27149e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +8.60639e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.73942e+03+1.27149e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.76006e+03+2.08250e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 523,521...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +6.54424e+03+7.20337e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -9.95787e+01+1.81480e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.17078e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +9.95787e+01+1.81480e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +6.54424e+03-7.20337e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 526,521...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.57498e+02+2.13320e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.94577e+02+1.40876e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.06630e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.94577e+02+1.40876e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.57498e+02-2.13320e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 529,521...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.11242e+02-2.17370e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -6.33389e+02-7.64456e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.62442e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +6.33389e+02-7.64456e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.11242e+02+2.17370e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 66,522...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.97289e+03+2.20731e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.42159e+02-1.97832e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.38487e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -4.42159e+02-1.97832e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.97289e+03-2.20731e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 68,522...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -6.93496e+02+3.47139e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.38907e+03-3.28246e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.39231e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.38907e+03-3.28246e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -6.93496e+02-3.47139e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 69,522...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.83461e+03-1.03197e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.85798e+03+1.03316e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.32547e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.85798e+03+1.03316e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.83461e+03+1.03197e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 70,522...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.12495e+03-2.86070e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.12435e+02-3.58277e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.83679e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +7.12435e+02-3.58277e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.12495e+03+2.86070e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 71,522...
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.42416e+03-2.04554e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.14960e+03-8.21375e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +7.97854e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.14960e+03-8.21375e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.42416e+03+2.04554e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 514,522...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.17813e+03-1.07200e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.67291e+02+4.32425e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.24546e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.67291e+02+4.32425e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.17813e+03+1.07200e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 521,522...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.8372 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 521,522...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.76006e+03-2.08250e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.73942e+03+1.27149e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +8.60639e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.73942e+03+1.27149e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.76006e+03+2.08250e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 523,522...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.70854e+03+2.40850e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.08042e+03-1.07491e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.65298e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.08042e+03-1.07491e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.70854e+03-2.40850e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 524,522...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +6.70797e+02+1.38413e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.90563e+02+3.04116e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.22168e+03 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.90563e+02+3.04116e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +6.70797e+02-1.38413e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 525,522...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.00685e+03+2.36332e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.51719e+03+3.27876e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +8.12873e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.51719e+03+3.27876e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.00685e+03-2.36332e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 526,522...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -6.74017e+02+1.66962e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.12817e+03+7.61197e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.05017e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.12817e+03+7.61197e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -6.74017e+02-1.66962e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 527,522...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.98431e+03-1.29330e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -4.08296e+02-3.19922e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.37979e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +4.08296e+02-3.19922e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.98431e+03+1.29330e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 528,522...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.15272e+03+8.56997e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +5.07628e+02-1.53362e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.31091e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -5.07628e+02-1.53362e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.15272e+03-8.56997e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 529,522...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.52333e+03+4.73545e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +7.00601e+02-4.61387e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.27100e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -7.00601e+02-4.61387e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.52333e+03-4.73545e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 530,522...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.67602e+01+7.08480e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.40537e+03+8.54091e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.37680e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.40537e+03+8.54091e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.67602e+01-7.08480e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 534,522...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.00616e+03+5.38001e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +5.94761e+02-2.37365e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.21095e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -5.94761e+02-2.37365e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.00616e+03-5.38001e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 312,523...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.20198e+02-1.50992e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -7.09745e+02-9.34232e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +9.21138e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +7.09745e+02-9.34232e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.20198e+02+1.50992e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 314,523...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -6.62723e+03-1.12249e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -6.48622e+03-5.45418e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.91486e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +6.48622e+03-5.45418e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -6.62723e+03+1.12249e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 315,523...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.75199e+03+9.99964e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.12185e+02-7.25649e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.36631e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -4.12185e+02-7.25649e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.75199e+03-9.99964e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 318,523...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.96475e+02+8.58587e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.23814e+03+8.82338e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.97249e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.23814e+03+8.82338e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.96475e+02-8.58587e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 339,523...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.84305e+02+4.36321e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.92075e+03-7.37621e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.11901e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.92075e+03-7.37621e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.84305e+02-4.36321e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 340,523...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.24627e+02+1.12629e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.10649e+03-1.73604e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.78065e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.10649e+03-1.73604e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.24627e+02-1.12629e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 341,523...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.92626e+02+6.11069e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.28498e+02-2.49900e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.80424e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.28498e+02-2.49900e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.92626e+02-6.11069e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 521,523...
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +6.54424e+03+7.20337e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -9.95787e+01+1.81480e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.17078e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +9.95787e+01+1.81480e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +6.54424e+03-7.20337e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 522,523...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
10.7004 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 522,523...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.70854e+03+2.40850e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.08042e+03-1.07491e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.65298e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.08042e+03-1.07491e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.70854e+03-2.40850e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 524,523...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
4.8085 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 524,523...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.14671e+03-1.39115e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.08333e+03+5.12966e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.52432e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.08333e+03+5.12966e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.14671e+03+1.39115e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 525,523...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.0786 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 525,523...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.40237e+02-9.33974e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.54729e+03+1.61540e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.75426e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.54729e+03+1.61540e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.40237e+02+9.33974e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 526,523...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
5.9449 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 526,523...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.71078e+02+3.37124e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.37610e+03+2.87230e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -9.21970e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.37610e+03+2.87230e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.71078e+02-3.37124e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 527,523...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.1123 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 527,523...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.38025e+03-2.28737e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +7.09661e+02+2.85714e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.62301e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -7.09661e+02+2.85714e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.38025e+03+2.28737e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 528,523...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
5.8544 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 528,523...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.02714e+03-6.14931e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.41707e+03-2.85429e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.17441e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.41707e+03-2.85429e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.02714e+03+6.14931e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 529,523...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
4.8823 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 529,523...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.72261e+03-8.17063e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -4.53640e+03-3.89530e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.14744e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +4.53640e+03-3.89530e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.72261e+03+8.17063e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 314,524...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.75331e+02-2.46163e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +6.63759e+02+5.93657e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.89175e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -6.63759e+02+5.93657e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.75331e+02+2.46163e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 318,524...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.94393e+03-5.39142e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.97267e+03-1.79145e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.90850e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.97267e+03-1.79145e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.94393e+03+5.39142e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 319,524...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.39059e+03+3.10933e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.69016e+02-1.86654e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.15516e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.69016e+02-1.86654e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.39059e+03-3.10933e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 522,524...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +6.70797e+02+1.38413e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.90563e+02+3.04116e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.22168e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.90563e+02+3.04116e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +6.70797e+02-1.38413e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 523,524...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.14671e+03-1.39115e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.08333e+03+5.12966e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.52432e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.08333e+03+5.12966e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.14671e+03+1.39115e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 525,524...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.43412e+02+1.16021e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.03951e+04-1.08000e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.57917e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.03951e+04-1.08000e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.43412e+02-1.16021e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 526,524...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.97892e+03-1.03656e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.86649e+03-1.19189e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.17270e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.86649e+03-1.19189e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.97892e+03+1.03656e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 527,524...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.59807e+03+2.24479e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.08916e+03+1.14970e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.51163e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.08916e+03+1.14970e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.59807e+03-2.24479e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 528,524...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.00548e+03+5.18922e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.54568e+03+1.10383e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -6.97112e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.54568e+03+1.10383e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.00548e+03-5.18922e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 529,524...
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.25292e+01+9.04314e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.31761e+03+1.29949e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -8.07507e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.31761e+03+1.29949e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.25292e+01-9.04314e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 69,525...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.61040e+02+2.22019e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.93047e+02-2.69138e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.90543e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -3.93047e+02-2.69138e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.61040e+02-2.22019e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 314,525...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.04107e+03-7.30526e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.03514e+03+3.26949e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +6.60164e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.03514e+03+3.26949e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.04107e+03+7.30526e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 318,525...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.33751e+03+5.85081e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +8.67858e+00-1.81514e+00i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.19197e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -8.67858e+00-1.81514e+00i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.33751e+03-5.85081e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 522,525...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.00685e+03+2.36332e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.51719e+03+3.27876e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +8.12873e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.51719e+03+3.27876e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.00685e+03-2.36332e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 523,525...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.40237e+02-9.33974e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.54729e+03+1.61540e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.75426e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.54729e+03+1.61540e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.40237e+02+9.33974e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 524,525...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 524,525...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.43412e+02+1.16021e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.03951e+04-1.08000e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.57917e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.03951e+04-1.08000e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.43412e+02-1.16021e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 526,525...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.50895e+04-6.26354e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.88525e+03+5.75037e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.31235e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.88525e+03+5.75037e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.50895e+04+6.26354e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 527,525...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.46222e+03-4.43060e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.89865e+03-7.25864e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.30525e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.89865e+03-7.25864e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.46222e+03+4.43060e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 528,525...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.55979e+02-7.29451e+00i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.85410e+01-1.22125e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.77402e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.85410e+01-1.22125e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.55979e+02+7.29451e+00i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 529,525...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.23755e+01+1.43956e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -7.08264e+02+6.49990e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.49314e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +7.08264e+02+6.49990e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.23755e+01-1.43956e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 530,525...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.37956e+03+4.17239e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.56289e+03-3.90628e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.74421e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.56289e+03-3.90628e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.37956e+03-4.17239e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 312,526...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.42463e+02-1.15556e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.63782e+02+6.66134e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +9.50258e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.63782e+02+6.66134e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.42463e+02+1.15556e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 314,526...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.19835e+03-3.40983e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +6.34108e+03+2.74620e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.52629e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -6.34108e+03+2.74620e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.19835e+03+3.40983e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 315,526...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.93991e+02-2.10906e+01i Hz
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```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.63996e+03+1.93419e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.97849e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.63996e+03+1.93419e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.93991e+02+2.10906e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 317,526...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.63537e+03-8.77816e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.45709e+02-6.17759e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.50136e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.45709e+02-6.17759e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.63537e+03+8.77816e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 318,526...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.41293e+03+5.14324e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -6.51834e+02+3.49750e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.00372e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +6.51834e+02+3.49750e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.41293e+03-5.14324e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 319,526...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -7.34094e+02+1.22817e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +6.88600e+02-3.90644e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +9.89442e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -6.88600e+02-3.90644e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -7.34094e+02-1.22817e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 521,526...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.57498e+02+2.13320e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.94577e+02+1.40876e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.06630e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.94577e+02+1.40876e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.57498e+02-2.13320e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 522,526...
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -6.74017e+02+1.66962e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.12817e+03+7.61197e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.05017e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.12817e+03+7.61197e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -6.74017e+02-1.66962e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 523,526...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.71078e+02+3.37124e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.37610e+03+2.87230e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -9.21970e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.37610e+03+2.87230e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.71078e+02-3.37124e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 524,526...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 524,526...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.97892e+03-1.03656e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.86649e+03-1.19189e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.17270e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.86649e+03-1.19189e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.97892e+03+1.03656e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 525,526...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 525,526...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.50895e+04-6.26354e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.88525e+03+5.75037e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.31235e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.88525e+03+5.75037e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.50895e+04+6.26354e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 527,526...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.20088e+03-4.39252e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.63127e+03-2.88977e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.67789e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.63127e+03-2.88977e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.20088e+03+4.39252e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 528,526...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.27435e+02-5.40981e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.41469e+03-9.39878e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.52765e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.41469e+03-9.39878e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.27435e+02+5.40981e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 530,526...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -9.06499e+02+2.28835e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.23523e+03-8.39299e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.63978e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.23523e+03-8.39299e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -9.06499e+02-2.28835e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 531,526...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.00833e+03+2.10897e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.37442e+02-1.32847e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.34218e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.37442e+02-1.32847e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.00833e+03-2.10897e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 543,526...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.38483e+02+5.52490e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.30703e+03+8.20192e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.22343e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.30703e+03+8.20192e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.38483e+02-5.52490e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 69,527...
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```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.65035e+02+9.58817e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.14956e+03-1.66771e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -7.94887e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.14956e+03-1.66771e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.65035e+02-9.58817e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 70,527...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.30294e+03+3.98640e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.58049e+03-7.42991e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.98644e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.58049e+03-7.42991e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.30294e+03-3.98640e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 71,527...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.68231e+02+1.36709e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.18719e+02+2.79408e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.13893e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.18719e+02+2.79408e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.68231e+02-1.36709e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 522,527...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.98431e+03-1.29330e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.08296e+02-3.19922e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.37979e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.08296e+02-3.19922e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.98431e+03+1.29330e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 523,527...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.38025e+03-2.28737e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +7.09661e+02+2.85714e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.62301e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -7.09661e+02+2.85714e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.38025e+03+2.28737e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 524,527...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.59807e+03+2.24479e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.08916e+03+1.14970e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.51163e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.08916e+03+1.14970e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.59807e+03-2.24479e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 525,527...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.46222e+03-4.43060e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.89865e+03-7.25864e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.30525e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.89865e+03-7.25864e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.46222e+03+4.43060e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 526,527...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.20088e+03-4.39252e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.63127e+03-2.88977e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.67789e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.63127e+03-2.88977e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.20088e+03+4.39252e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 528,527...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -6.84157e+03+8.43482e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.39898e+04-6.66591e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.59457e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.39898e+04-6.66591e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -6.84157e+03-8.43482e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 529,527...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.99949e+03-5.09549e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +7.81493e+03+1.36687e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.22890e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -7.81493e+03+1.36687e+04i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.99949e+03+5.09549e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 530,527...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +6.04632e+02-2.70464e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.49270e+02-1.63617e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.18436e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.49270e+02-1.63617e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +6.04632e+02+2.70464e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 340,528...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.75496e+03+6.79999e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.94018e+03-3.62744e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +6.91665e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.94018e+03-3.62744e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.75496e+03-6.79999e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 341,528...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.63354e+03-9.87300e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.76309e+03+1.32757e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.67093e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -4.76309e+03+1.32757e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.63354e+03+9.87300e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 522,528...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.15272e+03+8.56997e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.07628e+02-1.53362e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.31091e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.07628e+02-1.53362e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.15272e+03-8.56997e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 523,528...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.02714e+03-6.14931e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.41707e+03-2.85429e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.17441e+03 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.41707e+03-2.85429e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.02714e+03+6.14931e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 524,528...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.00548e+03+5.18922e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -4.54568e+03+1.10383e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -6.97112e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +4.54568e+03+1.10383e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.00548e+03-5.18922e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 525,528...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.55979e+02-7.29451e+00i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.85410e+01-1.22125e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.77402e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.85410e+01-1.22125e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.55979e+02+7.29451e+00i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 526,528...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.27435e+02-5.40981e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.41469e+03-9.39878e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.52765e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.41469e+03-9.39878e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.27435e+02+5.40981e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 527,528...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 527,528...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -6.84157e+03+8.43482e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.39898e+04-6.66591e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.59457e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.39898e+04-6.66591e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -6.84157e+03-8.43482e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 529,528...
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +9.36240e+03+1.25374e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.14658e+03+6.27684e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.14898e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.14658e+03+6.27684e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +9.36240e+03-1.25374e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 70,529...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.64635e+03+1.84084e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.15508e+03-1.20109e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.04529e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.15508e+03-1.20109e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.64635e+03-1.84084e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 71,529...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.10796e+03+8.07430e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.04977e+02-1.64481e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.03539e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.04977e+02-1.64481e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.10796e+03-8.07430e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 341,529...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.37500e+03+1.72903e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.60521e+03-1.25679e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.57554e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.60521e+03-1.25679e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.37500e+03-1.72903e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 507,529...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.97123e+00+2.80735e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -8.30155e+02+8.48000e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.81817e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +8.30155e+02+8.48000e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.97123e+00-2.80735e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 521,529...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.11242e+02-2.17370e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -6.33389e+02-7.64456e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.62442e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +6.33389e+02-7.64456e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.11242e+02+2.17370e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 522,529...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.52333e+03+4.73545e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +7.00601e+02-4.61387e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.27100e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -7.00601e+02-4.61387e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.52333e+03-4.73545e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 523,529...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.72261e+03-8.17063e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.53640e+03-3.89530e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.14744e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.53640e+03-3.89530e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.72261e+03+8.17063e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 524,529...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.25292e+01+9.04314e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.31761e+03+1.29949e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -8.07507e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.31761e+03+1.29949e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.25292e+01-9.04314e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 525,529...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.23755e+01+1.43956e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.08264e+02+6.49990e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.49314e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +7.08264e+02+6.49990e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.23755e+01-1.43956e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 527,529...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 527,529...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.99949e+03-5.09549e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +7.81493e+03+1.36687e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.22890e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -7.81493e+03+1.36687e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.99949e+03+5.09549e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 528,529...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 528,529...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +9.36240e+03+1.25374e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -3.14658e+03+6.27684e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.14898e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +3.14658e+03+6.27684e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +9.36240e+03-1.25374e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 66,530...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -7.81148e+02+9.02589e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.26284e+03+5.77178e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.15272e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.26284e+03+5.77178e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -7.81148e+02-9.02589e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 67,530...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.74156e+03-9.25192e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +6.31786e+01+1.57400e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.60930e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -6.31786e+01+1.57400e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.74156e+03+9.25192e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 68,530...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.40952e+03+1.33493e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +8.71202e+02-2.25207e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.12912e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -8.71202e+02-2.25207e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.40952e+03-1.33493e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 69,530...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.08432e+04-2.24190e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.09003e+03-5.20692e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +8.07543e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.09003e+03-5.20692e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.08432e+04+2.24190e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 70,530...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -6.05836e+02-6.40236e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.34316e+03-5.78196e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.70718e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.34316e+03-5.78196e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -6.05836e+02+6.40236e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 522,530...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.67602e+01+7.08480e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.40537e+03+8.54091e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.37680e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.40537e+03+8.54091e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.67602e+01-7.08480e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 525,530...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.37956e+03+4.17239e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.56289e+03-3.90628e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.74421e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.56289e+03-3.90628e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.37956e+03-4.17239e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 526,530...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -9.06499e+02+2.28835e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.23523e+03-8.39299e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.63978e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.23523e+03-8.39299e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -9.06499e+02-2.28835e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 527,530...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +6.04632e+02-2.70464e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.49270e+02-1.63617e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.18436e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.49270e+02-1.63617e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +6.04632e+02+2.70464e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 531,530...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -9.61036e+02-2.71087e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.10344e+03-1.48601e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.40694e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.10344e+03-1.48601e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -9.61036e+02+2.71087e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 532,530...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.72428e+02+7.82730e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.75338e+03-3.45451e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.52658e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.75338e+03-3.45451e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.72428e+02-7.82730e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 533,530...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.47027e+02-3.69141e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.91240e+02-1.60617e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.20638e+03 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +4.91240e+02-1.60617e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.47027e+02+3.69141e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 534,530...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.04684e+03+3.55743e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.83630e+02-1.21011e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.93501e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.83630e+02-1.21011e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.04684e+03-3.55743e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 539,530...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.37646e+03-1.41913e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +4.57643e+01+1.08143e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.61137e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -4.57643e+01+1.08143e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.37646e+03+1.41913e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 305,531...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.05705e+03-1.89809e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.13621e+03+1.25547e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +6.20214e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.13621e+03+1.25547e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.05705e+03+1.89809e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 312,531...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.91829e+02-1.50386e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.45646e+03+2.22618e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.35862e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.45646e+03+2.22618e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.91829e+02+1.50386e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 526,531...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.00833e+03+2.10897e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.37442e+02-1.32847e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.34218e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.37442e+02-1.32847e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.00833e+03-2.10897e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 530,531...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.6346 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 530,531...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -9.61036e+02-2.71087e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.10344e+03-1.48601e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.40694e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.10344e+03-1.48601e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -9.61036e+02+2.71087e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 532,531...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.59847e+03-9.86329e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.26621e+03+4.15636e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.48961e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.26621e+03+4.15636e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.59847e+03+9.86329e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 533,531...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.20732e+03+2.25068e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +6.01925e+03-2.52761e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.94549e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -6.01925e+03-2.52761e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.20732e+03-2.25068e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 534,531...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.22605e+03+1.20786e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.55712e+03+3.95233e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.65187e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.55712e+03+3.95233e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.22605e+03-1.20786e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 538,531...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +8.04325e+02+1.44943e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.18257e+02+7.10443e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.18607e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.18257e+02+7.10443e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +8.04325e+02-1.44943e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 539,531...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.08334e+03+3.78720e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.89364e+03+6.08723e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.89947e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.89364e+03+6.08723e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.08334e+03-3.78720e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 540,531...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.47783e+03+3.12486e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.05267e+02-1.28890e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.32189e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -4.05267e+02-1.28890e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.47783e+03-3.12486e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 541,531...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.21796e+03+3.38181e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.06836e+03+4.96713e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.54189e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +7.06836e+03+4.96713e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.21796e+03-3.38181e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 68,532...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.86587e+03-4.16885e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.81814e+03-3.10991e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.55484e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.81814e+03-3.10991e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.86587e+03+4.16885e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 69,532...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -6.75976e+01-7.10333e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.59457e+03-1.45003e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.07513e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.59457e+03-1.45003e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -6.75976e+01+7.10333e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 530,532...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.72428e+02+7.82730e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.75338e+03-3.45451e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.52658e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.75338e+03-3.45451e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.72428e+02-7.82730e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 531,532...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
12.5891 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 531,532...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.59847e+03-9.86329e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.26621e+03+4.15636e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.48961e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.26621e+03+4.15636e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.59847e+03+9.86329e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 533,532...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -7.85223e+02-1.47031e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.25558e+03-6.87843e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +9.25069e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +7.25558e+03-6.87843e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -7.85223e+02+1.47031e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 534,532...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
5.3126 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 534,532...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.49586e+03+4.77946e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -8.10331e+02+5.19859e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.91434e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +8.10331e+02+5.19859e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.49586e+03-4.77946e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 535,532...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.57090e+02+5.90295e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.71459e+02+6.49434e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.80789e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.71459e+02+6.49434e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.57090e+02-5.90295e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 536,532...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.15304e+03+1.09954e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.10378e+02+1.02500e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +9.88542e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.10378e+02+1.02500e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.15304e+03-1.09954e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 537,532...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.89512e+03+8.33738e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +8.59185e+01-6.08823e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.40870e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -8.59185e+01-6.08823e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.89512e+03-8.33738e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 539,532...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.33408e+02-9.16396e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.01794e+03+1.17687e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.11265e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.01794e+03+1.17687e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.33408e+02+9.16396e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 540,532...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +6.42955e+02-1.67381e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.88531e+02+7.10991e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.29458e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -4.88531e+02+7.10991e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +6.42955e+02+1.67381e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 541,532...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.44267e+03+2.20363e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.82460e+02+2.90414e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.15044e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.82460e+02+2.90414e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.44267e+03-2.20363e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 530,533...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.47027e+02-3.69141e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.91240e+02-1.60617e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.20638e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.91240e+02-1.60617e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.47027e+02+3.69141e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 531,533...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
4.6709 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 531,533...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.20732e+03+2.25068e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +6.01925e+03-2.52761e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.94549e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -6.01925e+03-2.52761e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.20732e+03-2.25068e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 532,533...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
```

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[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 532,533...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -7.85223e+02-1.47031e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.25558e+03-6.87843e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +9.25069e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +7.25558e+03-6.87843e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -7.85223e+02+1.47031e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 534,533...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.2198 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 534,533...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -6.88166e+02+1.00304e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.64330e+03+1.91625e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.55540e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.64330e+03+1.91625e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -6.88166e+02-1.00304e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 535,533...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.72520e+03+1.43087e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.52490e+03+3.75988e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.18601e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.52490e+03+3.75988e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.72520e+03-1.43087e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 536,533...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.42736e+02+1.32871e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.12810e+03+9.40685e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.50739e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.12810e+03+9.40685e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.42736e+02-1.32871e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 537,533...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.99538e+02+5.49914e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.33026e+02+8.47310e+02i Hz
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```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.44502e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +7.33026e+02+8.47310e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.99538e+02-5.49914e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 538,533...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +8.81017e+02+2.12141e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.26025e+02+1.90418e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -9.16606e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.26025e+02+1.90418e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +8.81017e+02-2.12141e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 539,533...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.48817e+02-3.52055e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +9.91947e+02+1.49623e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.13009e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -9.91947e+02+1.49623e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.48817e+02+3.52055e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 540,533...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.87908e+03-3.20683e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.58668e+03+5.79853e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.00123e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.58668e+03+5.79853e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.87908e+03+3.20683e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 541,533...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.75973e+03+2.68479e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.02260e+02-5.52426e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.61932e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.02260e+02-5.52426e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.75973e+03-2.68479e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 553,533...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.95828e+02+5.94898e+01i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -9.56197e+01+1.27960e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.35250e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +9.56197e+01+1.27960e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.95828e+02-5.94898e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 66,534...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.01699e+03-9.94899e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.32642e+03-9.48705e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -6.49684e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.32642e+03-9.48705e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.01699e+03+9.94899e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 68,534...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.62798e+03-3.87201e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +6.64787e+02+4.41648e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.36764e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -6.64787e+02+4.41648e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.62798e+03+3.87201e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 69,534...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.30920e+03-2.81086e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.78738e+02-1.64758e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.59753e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +7.78738e+02-1.64758e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.30920e+03+2.81086e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 514,534...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.24161e+02-5.90109e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.33323e+03+9.19730e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.18121e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.33323e+03+9.19730e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.24161e+02+5.90109e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 522,534...
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[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.00616e+03+5.38001e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.94761e+02-2.37365e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.21095e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.94761e+02-2.37365e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.00616e+03-5.38001e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 530,534...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.04684e+03+3.55743e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.83630e+02-1.21011e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.93501e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -3.83630e+02-1.21011e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.04684e+03-3.55743e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 531,534...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.22605e+03+1.20786e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.55712e+03+3.95233e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.65187e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.55712e+03+3.95233e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.22605e+03-1.20786e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 532,534...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.49586e+03+4.77946e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -8.10331e+02+5.19859e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.91434e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +8.10331e+02+5.19859e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.49586e+03-4.77946e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 533,534...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -6.88166e+02+1.00304e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.64330e+03+1.91625e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.55540e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.64330e+03+1.91625e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -6.88166e+02-1.00304e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 535,534...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.60672e+03+1.62537e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.27838e+03-2.20424e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.97861e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.27838e+03-2.20424e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.60672e+03-1.62537e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 536,534...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.79457e+03+4.46744e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.04412e+03-3.68979e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.92404e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.04412e+03-3.68979e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.79457e+03-4.46744e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 537,534...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.10952e+03+2.95410e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.00515e+02-3.06095e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.45099e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -4.00515e+02-3.06095e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.10952e+03-2.95410e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 538,534...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.21606e+03-3.91740e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.11309e+02-2.35395e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.13290e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.11309e+02-2.35395e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.21606e+03+3.91740e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 539,534...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.33038e+03-5.06452e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.83835e+03+8.46869e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.18969e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.83835e+03+8.46869e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.33038e+03+5.06452e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 540,534...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -7.36810e+02-2.57789e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.98924e+03-1.50034e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.24384e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.98924e+03-1.50034e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -7.36810e+02+2.57789e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 64,535...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.80657e+02-5.45518e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.30233e+03-5.15798e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.54904e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.30233e+03-5.15798e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.80657e+02+5.45518e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 68,535...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.82571e+03-1.78271e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.41006e+03-3.46238e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.52588e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.41006e+03-3.46238e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.82571e+03+1.78271e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 532,535...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.57090e+02+5.90295e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.71459e+02+6.49434e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.80789e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.71459e+02+6.49434e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.57090e+02-5.90295e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 533,535...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.72520e+03+1.43087e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.52490e+03+3.75988e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.18601e+03 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.52490e+03+3.75988e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.72520e+03-1.43087e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 534,535...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
5.157 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 534,535...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.60672e+03+1.62537e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +5.27838e+03-2.20424e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.97861e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -5.27838e+03-2.20424e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.60672e+03-1.62537e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 536,535...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +9.66442e+03-4.83479e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.57837e+03+1.51510e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.32623e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.57837e+03+1.51510e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +9.66442e+03+4.83479e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 537,535...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.26390e+01-1.51352e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -6.22841e+03-6.23362e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.02633e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +6.22841e+03-6.23362e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.26390e+01+1.51352e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 538,535...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.91602e+02-1.30028e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.38524e+03+9.57211e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.02388e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.38524e+03+9.57211e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.91602e+02+1.30028e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 540,535...
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.31260e+03-8.33942e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.29034e+02+1.53845e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.19006e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.29034e+02+1.53845e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.31260e+03+8.33942e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 68,536...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.29386e+03-9.92960e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.83233e+02-5.39724e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.25036e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.83233e+02-5.39724e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.29386e+03+9.92960e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 532,536...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.15304e+03+1.09954e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.10378e+02+1.02500e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +9.88542e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.10378e+02+1.02500e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.15304e+03-1.09954e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 533,536...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.42736e+02+1.32871e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.12810e+03+9.40685e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.50739e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.12810e+03+9.40685e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.42736e+02-1.32871e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 534,536...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
5.4651 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 534,536...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.79457e+03+4.46744e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.04412e+03-3.68979e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.92404e+03 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.04412e+03-3.68979e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.79457e+03-4.46744e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 535,536...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 535,536...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +9.66442e+03-4.83479e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.57837e+03+1.51510e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.32623e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.57837e+03+1.51510e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +9.66442e+03+4.83479e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 537,536...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.76635e+03+1.78220e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.39286e+04+3.12912e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.65655e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.39286e+04+3.12912e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.76635e+03-1.78220e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 538,536...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.39827e+03-1.83673e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +4.22398e+03+6.98918e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.34344e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -4.22398e+03+6.98918e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.39827e+03+1.83673e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 539,536...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.37741e+03+9.59668e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.19935e+03-3.76609e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +9.86397e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.19935e+03-3.76609e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.37741e+03-9.59668e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 540,536...
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.36122e+03+4.92996e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.26346e+02+5.43613e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.92211e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.26346e+02+5.43613e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.36122e+03-4.92996e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 532,537...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.89512e+03+8.33738e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +8.59185e+01-6.08823e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.40870e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -8.59185e+01-6.08823e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.89512e+03-8.33738e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 533,537...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.99538e+02+5.49914e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.33026e+02+8.47310e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.44502e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +7.33026e+02+8.47310e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.99538e+02-5.49914e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 534,537...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.2097 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 534,537...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.10952e+03+2.95410e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.00515e+02-3.06095e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.45099e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -4.00515e+02-3.06095e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.10952e+03-2.95410e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 535,537...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 535,537...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.26390e+01-1.51352e+04i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -6.22841e+03-6.23362e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.02633e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +6.22841e+03-6.23362e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.26390e+01+1.51352e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 536,537...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 536,537...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.76635e+03+1.78220e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.39286e+04+3.12912e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.65655e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.39286e+04+3.12912e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.76635e+03-1.78220e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 538,537...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.12888e+02-8.18685e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.44955e+03+1.43603e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.99843e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.44955e+03+1.43603e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.12888e+02+8.18685e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 539,537...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.99406e+02+4.72767e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.61249e+03-5.59381e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -9.34482e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.61249e+03-5.59381e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.99406e+02-4.72767e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 540,537...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.73364e+03+4.48285e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.58937e+03-2.14711e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.66670e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.58937e+03-2.14711e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.73364e+03-4.48285e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 272,538...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.21871e+01-8.64857e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.08886e+03-1.02513e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.46372e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.08886e+03-1.02513e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.21871e+01+8.64857e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 274,538...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.07134e+03-2.37511e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.26730e+03-8.18620e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.77079e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.26730e+03-8.18620e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.07134e+03+2.37511e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 277,538...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.89705e+03-2.12437e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.00702e+02+7.87508e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.56324e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.00702e+02+7.87508e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.89705e+03+2.12437e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 308,538...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.79407e+03-5.22311e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +9.72617e+02+1.38701e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.31479e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -9.72617e+02+1.38701e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.79407e+03+5.22311e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 514,538...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +9.97675e+01+3.71882e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.09059e+03-1.42174e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.08994e+03 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.09059e+03-1.42174e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +9.97675e+01-3.71882e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 518,538...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.11738e+02-1.42116e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.53575e+03+3.37469e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.70836e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.53575e+03+3.37469e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.11738e+02+1.42116e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 519,538...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.63598e+02+2.91779e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -8.69933e+02+1.48511e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.34218e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +8.69933e+02+1.48511e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.63598e+02-2.91779e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 531,538...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +8.04325e+02+1.44943e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -4.18257e+02+7.10443e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.18607e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +4.18257e+02+7.10443e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +8.04325e+02-1.44943e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 533,538...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +8.81017e+02+2.12141e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.26025e+02+1.90418e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -9.16606e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.26025e+02+1.90418e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +8.81017e+02-2.12141e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 534,538...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
4.548 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 534,538...
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```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.21606e+03-3.91740e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.11309e+02-2.35395e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.13290e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.11309e+02-2.35395e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.21606e+03+3.91740e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 535,538...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.91602e+02-1.30028e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.38524e+03+9.57211e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.02388e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.38524e+03+9.57211e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.91602e+02+1.30028e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 536,538...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.39827e+03-1.83673e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.22398e+03+6.98918e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.34344e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -4.22398e+03+6.98918e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.39827e+03+1.83673e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 537,538...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.12888e+02-8.18685e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.44955e+03+1.43603e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.99843e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -3.44955e+03+1.43603e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.12888e+02+8.18685e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 539,538...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.73543e+03+1.48304e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.53508e+03-6.20875e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.11610e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -4.53508e+03-6.20875e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.73543e+03-1.48304e+04i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 540,538...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.45180e+03+8.01992e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.36917e+04+9.01307e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.43463e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.36917e+04+9.01307e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.45180e+03-8.01992e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 274,539...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +9.47818e+02-1.31264e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.64354e+02-1.10360e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +9.34554e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.64354e+02-1.10360e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +9.47818e+02+1.31264e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 305,539...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.65470e+03+5.70781e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.10354e+02-4.36163e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.19149e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -4.10354e+02-4.36163e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.65470e+03-5.70781e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 308,539...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -7.42616e+02-2.91981e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +7.55772e+02+5.87612e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.33573e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -7.55772e+02+5.87612e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -7.42616e+02+2.91981e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 312,539...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.94957e+02+9.97695e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.10737e+03-6.28960e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.78465e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.10737e+03-6.28960e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.94957e+02-9.97695e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 514,539...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.89630e+01+9.81866e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.55836e+02+1.22152e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -7.58807e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +7.55836e+02+1.22152e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.89630e+01-9.81866e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 515,539...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.28261e+02-7.91828e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.30601e+03+1.11149e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -8.41890e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.30601e+03+1.11149e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.28261e+02+7.91828e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 518,539...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.88960e+03+1.02215e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.01537e+01+2.59569e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +9.05097e+01 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +7.01537e+01+2.59569e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.88960e+03-1.02215e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 519,539...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.13228e+02+7.57877e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -8.30641e+02+1.24209e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.41960e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +8.30641e+02+1.24209e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.13228e+02-7.57877e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 530,539...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.37646e+03-1.41913e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.57643e+01+1.08143e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.61137e+03 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -4.57643e+01+1.08143e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.37646e+03+1.41913e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 531,539...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.08334e+03+3.78720e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.89364e+03+6.08723e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.89947e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.89364e+03+6.08723e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.08334e+03-3.78720e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 532,539...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.33408e+02-9.16396e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.01794e+03+1.17687e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.11265e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.01794e+03+1.17687e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.33408e+02+9.16396e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 533,539...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.48817e+02-3.52055e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +9.91947e+02+1.49623e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.13009e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -9.91947e+02+1.49623e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.48817e+02+3.52055e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 534,539...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
6.5779 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 534,539...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.33038e+03-5.06452e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.83835e+03+8.46869e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.18969e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.83835e+03+8.46869e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.33038e+03+5.06452e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 536,539...
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.37741e+03+9.59668e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.19935e+03-3.76609e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +9.86397e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.19935e+03-3.76609e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.37741e+03-9.59668e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 537,539...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.99406e+02+4.72767e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.61249e+03-5.59381e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -9.34482e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.61249e+03-5.59381e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.99406e+02-4.72767e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 538,539...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 538,539...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.73543e+03+1.48304e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.53508e+03-6.20875e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.11610e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -4.53508e+03-6.20875e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.73543e+03-1.48304e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 540,539...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.36748e+03-4.68859e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.13488e+03-1.15723e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.09002e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.13488e+03-1.15723e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.36748e+03+4.68859e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 272,540...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.61123e+03-1.50641e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.41082e+02-1.87777e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.04673e+03 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +7.41082e+02-1.87777e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.61123e+03+1.50641e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 274,540...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.73862e+03-3.62217e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +4.98816e+02+7.92731e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.19140e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -4.98816e+02+7.92731e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.73862e+03+3.62217e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 277,540...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.50390e+03+3.71075e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.62039e+02+1.33312e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +7.89437e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.62039e+02+1.33312e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.50390e+03-3.71075e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 305,540...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.02430e+03+7.43469e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.37663e+03-4.46941e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.57711e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.37663e+03-4.46941e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.02430e+03-7.43469e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 308,540...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.38813e+02-1.16077e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.05494e+03+1.05036e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.48933e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.05494e+03+1.05036e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.38813e+02+1.16077e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 518,540...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.55219e+02+5.00567e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -8.79195e+01+1.25398e+03i Hz
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```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.50542e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +8.79195e+01+1.25398e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.55219e+02-5.00567e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 531,540...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.47783e+03+3.12486e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +4.05267e+02-1.28890e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.32189e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -4.05267e+02-1.28890e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.47783e+03-3.12486e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 532,540...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +6.42955e+02-1.67381e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +4.88531e+02+7.10991e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.29458e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -4.88531e+02+7.10991e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +6.42955e+02+1.67381e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 533,540...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.87908e+03-3.20683e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.58668e+03+5.79853e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.00123e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.58668e+03+5.79853e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.87908e+03+3.20683e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 534,540...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
8.7165 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 534,540...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -7.36810e+02-2.57789e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.98924e+03-1.50034e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.24384e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.98924e+03-1.50034e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -7.36810e+02+2.57789e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 535,540...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.31260e+03-8.33942e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.29034e+02+1.53845e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.19006e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.29034e+02+1.53845e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.31260e+03+8.33942e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 536,540...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.36122e+03+4.92996e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.26346e+02+5.43613e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.92211e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.26346e+02+5.43613e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.36122e+03-4.92996e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 537,540...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.73364e+03+4.48285e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.58937e+03-2.14711e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.66670e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -3.58937e+03-2.14711e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.73364e+03-4.48285e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 538,540...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 538,540...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.45180e+03+8.01992e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.36917e+04+9.01307e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.43463e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.36917e+04+9.01307e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.45180e+03-8.01992e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 539,540...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 539,540...
```

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[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.36748e+03-4.68859e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.13488e+03-1.15723e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.09002e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.13488e+03-1.15723e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.36748e+03+4.68859e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 541,540...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +6.18489e+02+1.12540e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.78390e+02+8.08786e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +7.67791e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.78390e+02+8.08786e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +6.18489e+02-1.12540e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 297,541...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.36940e+03-1.24044e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.70607e+02-2.97130e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.35792e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +7.70607e+02-2.97130e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.36940e+03+1.24044e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 299,541...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.45531e+03-1.74354e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -9.04430e+02-1.93300e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.03563e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +9.04430e+02-1.93300e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.45531e+03+1.74354e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 531,541...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.21796e+03+3.38181e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.06836e+03+4.96713e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.54189e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +7.06836e+03+4.96713e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.21796e+03-3.38181e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 532,541...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.44267e+03+2.20363e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.82460e+02+2.90414e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.15044e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.82460e+02+2.90414e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.44267e+03-2.20363e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 533,541...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.75973e+03+2.68479e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.02260e+02-5.52426e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.61932e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.02260e+02-5.52426e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.75973e+03-2.68479e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 540,541...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +6.18489e+02+1.12540e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.78390e+02+8.08786e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +7.67791e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.78390e+02+8.08786e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +6.18489e+02-1.12540e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 542,541...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.85509e+03-1.26868e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.04916e+02+1.90840e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.05364e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -4.04916e+02+1.90840e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.85509e+03+1.26868e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 543,541...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.68581e+03+2.60668e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.00182e+02-9.19150e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.39065e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.00182e+02-9.19150e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.68581e+03-2.60668e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 544,541...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.92845e+03+4.64851e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.86940e+03+1.24836e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.69924e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.86940e+03+1.24836e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.92845e+03-4.64851e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 541,542...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.8659 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 541,542...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.85509e+03-1.26868e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.04916e+02+1.90840e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.05364e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -4.04916e+02+1.90840e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.85509e+03+1.26868e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 543,542...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.07399e+03-5.62222e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.63031e+03+3.86666e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.96267e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.63031e+03+3.86666e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.07399e+03+5.62222e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 544,542...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.85525e+03-2.60663e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +6.13435e+02+3.16338e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.55155e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -6.13435e+02+3.16338e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.85525e+03+2.60663e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 545,542...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.17268e+03+2.69752e+02i Hz
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[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.43887e+03+1.50822e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +6.74230e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.43887e+03+1.50822e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.17268e+03-2.69752e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 546,542...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.73785e+03+5.51346e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.31918e+03+1.70065e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.13596e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.31918e+03+1.70065e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.73785e+03-5.51346e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 549,542...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.85156e+01-2.84634e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.27070e+03+2.12779e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.36268e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.27070e+03+2.12779e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.85156e+01+2.84634e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 553,542...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +8.46793e+01+8.90211e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.06960e+03-1.17617e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.23720e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.06960e+03-1.17617e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +8.46793e+01-8.90211e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 317,543...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.09243e+02-4.68122e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.39798e+03+1.10930e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.31237e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.39798e+03+1.10930e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.09243e+02+4.68122e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 318,543...
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.25509e+02-3.76961e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.17931e+02-1.42078e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.44253e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.17931e+02-1.42078e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.25509e+02+3.76961e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 526,543...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.38483e+02+5.52490e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.30703e+03+8.20192e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.22343e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.30703e+03+8.20192e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.38483e+02-5.52490e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 541,543...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.68581e+03+2.60668e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.00182e+02-9.19150e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.39065e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.00182e+02-9.19150e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.68581e+03-2.60668e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 542,543...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
3.3933 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 542,543...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.07399e+03-5.62222e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.63031e+03+3.86666e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.96267e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.63031e+03+3.86666e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.07399e+03+5.62222e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 544,543...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.89418e+03-8.58889e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.37870e+03-1.40128e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.32395e+03 Hz
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```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +7.37870e+03-1.40128e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.89418e+03+8.58889e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 545,543...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.148 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 545,543...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.50940e+02+1.57224e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.49032e+03+6.35184e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.73808e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.49032e+03+6.35184e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.50940e+02-1.57224e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 546,543...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
4.9473 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 546,543...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.12176e+03-2.98286e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.83096e+02+5.47686e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.99385e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.83096e+02+5.47686e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.12176e+03+2.98286e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 547,543...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.74671e+01+5.36063e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.46420e+03+1.47173e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.04877e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.46420e+03+1.47173e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.74671e+01-5.36063e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 548,543...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.21326e+03+8.82511e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -3.00002e+03+4.04484e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.59807e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +3.00002e+03+4.04484e+02i Hz
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```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.21326e+03-8.82511e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 297,544...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.41182e+03-2.68208e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.25105e+02-5.05155e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.91049e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.25105e+02-5.05155e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.41182e+03+2.68208e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 541,544...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.92845e+03+4.64851e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.86940e+03+1.24836e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.69924e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.86940e+03+1.24836e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.92845e+03-4.64851e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 542,544...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
12.8996 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 542,544...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.85525e+03-2.60663e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +6.13435e+02+3.16338e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.55155e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -6.13435e+02+3.16338e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.85525e+03+2.60663e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 543,544...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 543,544...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.89418e+03-8.58889e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -7.37870e+03-1.40128e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.32395e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +7.37870e+03-1.40128e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.89418e+03+8.58889e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 545,544...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
5.7325 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 545,544...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.58211e+03+2.36899e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.84401e+02+6.21275e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.97271e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.84401e+02+6.21275e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.58211e+03-2.36899e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 546,544...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.1569 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 546,544...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.53273e+03-1.49748e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +4.92352e+02+1.80012e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.91914e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -4.92352e+02+1.80012e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.53273e+03+1.49748e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 547,544...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.93903e+03+1.63677e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.19396e+02-8.73539e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.93267e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.19396e+02-8.73539e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.93903e+03-1.63677e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 548,544...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.04149e+03+5.88738e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -3.67729e+02+3.08386e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.86593e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +3.67729e+02+3.08386e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.04149e+03-5.88738e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 542,545...
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.17268e+03+2.69752e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.43887e+03+1.50822e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +6.74230e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.43887e+03+1.50822e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.17268e+03-2.69752e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 543,545...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.50940e+02+1.57224e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.49032e+03+6.35184e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.73808e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.49032e+03+6.35184e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.50940e+02-1.57224e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 544,545...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.58211e+03+2.36899e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.84401e+02+6.21275e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.97271e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.84401e+02+6.21275e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.58211e+03-2.36899e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 546,545...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.37671e+03-1.08321e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -8.29732e+03-1.23015e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.84531e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +8.29732e+03-1.23015e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.37671e+03+1.08321e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 547,545...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.74207e+02+1.05929e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.78590e+03-2.52496e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.55850e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.78590e+03-2.52496e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.74207e+02-1.05929e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 548,545...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.85548e+03+1.06784e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.53707e+03-1.03497e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.04870e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.53707e+03-1.03497e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.85548e+03-1.06784e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 549,545...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.91677e+03-1.88561e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.15638e+03+3.73395e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.19953e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.15638e+03+3.73395e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.91677e+03+1.88561e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 550,545...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.28802e+01+9.20346e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.08266e+03-1.02224e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.29150e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.08266e+03-1.02224e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.28802e+01-9.20346e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 542,546...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.73785e+03+5.51346e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.31918e+03+1.70065e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.13596e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.31918e+03+1.70065e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.73785e+03-5.51346e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 543,546...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.12176e+03-2.98286e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.83096e+02+5.47686e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.99385e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -3.83096e+02+5.47686e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.12176e+03+2.98286e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 544,546...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.53273e+03-1.49748e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.92352e+02+1.80012e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.91914e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -4.92352e+02+1.80012e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.53273e+03+1.49748e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 545,546...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 545,546...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.37671e+03-1.08321e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -8.29732e+03-1.23015e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.84531e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +8.29732e+03-1.23015e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.37671e+03+1.08321e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 547,546...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.85647e+03+9.12259e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +6.31435e+03-7.36674e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -9.27628e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -6.31435e+03-7.36674e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.85647e+03-9.12259e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 548,546...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.99262e+03-2.27041e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.54449e+03+6.99444e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.07799e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.54449e+03+6.99444e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.99262e+03+2.27041e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 549,546...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.50545e+03+2.11986e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.88428e+03-1.48930e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.68429e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.88428e+03-1.48930e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.50545e+03-2.11986e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 543,547...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.74671e+01+5.36063e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.46420e+03+1.47173e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.04877e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.46420e+03+1.47173e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.74671e+01-5.36063e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 544,547...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.93903e+03+1.63677e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.19396e+02-8.73539e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.93267e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -3.19396e+02-8.73539e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.93903e+03-1.63677e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 545,547...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.0867 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 545,547...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.74207e+02+1.05929e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.78590e+03-2.52496e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.55850e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.78590e+03-2.52496e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.74207e+02-1.05929e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 546,547...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
5.9269 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 546,547...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.85647e+03+9.12259e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +6.31435e+03-7.36674e+02i Hz
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```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -9.27628e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -6.31435e+03-7.36674e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.85647e+03-9.12259e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 548,547...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.32350e+03-1.18901e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -5.74378e+03-1.02837e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +7.34577e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +5.74378e+03-1.02837e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.32350e+03+1.18901e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 543,548...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.21326e+03+8.82511e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -3.00002e+03+4.04484e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.59807e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +3.00002e+03+4.04484e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.21326e+03-8.82511e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 544,548...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.04149e+03+5.88738e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -3.67729e+02+3.08386e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.86593e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +3.67729e+02+3.08386e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.04149e+03-5.88738e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 545,548...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
4.7966 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 545,548...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.85548e+03+1.06784e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +5.53707e+03-1.03497e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.04870e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -5.53707e+03-1.03497e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.85548e+03-1.06784e+02i Hz
```

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[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 546,548...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.0628 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 546,548...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.99262e+03-2.27041e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.54449e+03+6.99444e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.07799e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.54449e+03+6.99444e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.99262e+03+2.27041e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 547,548...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 547,548...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.32350e+03-1.18901e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -5.74378e+03-1.02837e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +7.34577e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +5.74378e+03-1.02837e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.32350e+03+1.18901e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 542,549...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.85156e+01-2.84634e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.27070e+03+2.12779e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.36268e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.27070e+03+2.12779e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.85156e+01+2.84634e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 545,549...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.91677e+03-1.88561e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.15638e+03+3.73395e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.19953e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.15638e+03+3.73395e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.91677e+03+1.88561e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 546,549...
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.50545e+03+2.11986e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.88428e+03-1.48930e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.68429e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.88428e+03-1.48930e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.50545e+03-2.11986e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 550,549...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.56884e+03-1.40027e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.18692e+03-3.11227e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.37000e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.18692e+03-3.11227e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.56884e+03+1.40027e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 551,549...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.03758e+03+9.79211e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.27205e+03-1.20106e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.43403e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.27205e+03-1.20106e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.03758e+03-9.79211e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 552,549...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.09184e+02+4.06176e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +9.07117e+02-1.18316e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.81407e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -9.07117e+02-1.18316e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.09184e+02-4.06176e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 553,549...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.26384e+03+3.29499e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.63199e+02-8.19230e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.76713e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.63199e+02-8.19230e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.26384e+03-3.29499e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 545,550...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.28802e+01+9.20346e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.08266e+03-1.02224e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.29150e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.08266e+03-1.02224e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.28802e+01-9.20346e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 549,550...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x 9.87↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 549,550...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.56884e+03-1.40027e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.18692e+03-3.11227e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.37000e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.18692e+03-3.11227e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.56884e+03+1.40027e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 551,550...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -9.04345e+02-3.11285e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.74745e+02+1.31203e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.64071e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.74745e+02+1.31203e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -9.04345e+02+3.11285e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 552,550...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.98635e+03-2.59029e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.98792e+03+1.86182e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.00524e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -4.98792e+03+1.86182e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.98635e+03+2.59029e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 553,550...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.81762e+03+4.77543e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.30377e+03+2.77987e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +7.63104e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +3.30377e+03+2.77987e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.81762e+03-4.77543e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 557,550...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -7.99367e+02+3.12766e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.45373e+03-1.90548e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.41523e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.45373e+03-1.90548e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -7.99367e+02-3.12766e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 558,550...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.40305e+02+1.64494e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.83253e+01+2.30652e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.37120e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.83253e+01+2.30652e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.40305e+02-1.64494e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 559,550...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.84528e+03+6.65013e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.14682e+03+1.73804e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.66120e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.14682e+03+1.73804e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.84528e+03-6.65013e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 560,550...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.93466e+03+3.00462e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -8.23390e+03+3.46754e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.32949e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +8.23390e+03+3.46754e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.93466e+03-3.00462e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 564,550...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.33581e+03+6.50393e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -8.86250e+02+2.04289e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +9.85807e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +8.86250e+02+2.04289e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.33581e+03-6.50393e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 568,550...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.60112e+02+4.33267e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.25730e+02+9.45966e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.10593e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.25730e+02+9.45966e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.60112e+02-4.33267e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 549,551...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.03758e+03+9.79211e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.27205e+03-1.20106e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.43403e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.27205e+03-1.20106e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.03758e+03-9.79211e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 550,551...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
12.8159 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 550,551...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -9.04345e+02-3.11285e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.74745e+02+1.31203e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.64071e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.74745e+02+1.31203e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -9.04345e+02+3.11285e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 552,551...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.15485e+03-5.53850e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -6.57919e+03-1.51114e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -8.47899e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +6.57919e+03-1.51114e+04i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.15485e+03+5.53850e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 553,551...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
4.766 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 553,551...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.45651e+03+1.77637e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.35179e+02+5.26420e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.44922e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.35179e+02+5.26420e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.45651e+03-1.77637e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 554,551...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.34791e+02+3.73924e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.69507e+03-1.63506e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.44986e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.69507e+03-1.63506e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.34791e+02-3.73924e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 555,551...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.25853e+03+4.36876e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.98395e+03+1.49692e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.61522e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.98395e+03+1.49692e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.25853e+03-4.36876e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 556,551...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.89983e+02+1.75024e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -3.73807e+02+4.66265e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.38280e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +3.73807e+02+4.66265e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.89983e+02-1.75024e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 557,551...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.71089e+03-4.16908e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.38723e+02-1.15522e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.12396e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.38723e+02-1.15522e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.71089e+03+4.16908e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 559,551...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.21275e+03-8.42476e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +9.74092e+01+3.10957e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.17633e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -9.74092e+01+3.10957e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.21275e+03+8.42476e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 549,552...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.09184e+02+4.06176e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +9.07117e+02-1.18316e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.81407e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -9.07117e+02-1.18316e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.09184e+02-4.06176e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 550,552...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
2.7355 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 550,552...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.98635e+03-2.59029e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.98792e+03+1.86182e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.00524e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -4.98792e+03+1.86182e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.98635e+03+2.59029e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 551,552...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 551,552...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.15485e+03-5.53850e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -6.57919e+03-1.51114e+04i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -8.47899e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +6.57919e+03-1.51114e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.15485e+03+5.53850e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 553,552...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.0551 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 553,552...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.08743e+02+3.45585e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.04971e+03+1.86039e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.28428e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.04971e+03+1.86039e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.08743e+02-3.45585e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 554,552...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.26737e+03+3.01973e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.80708e+03+5.74706e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.98762e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.80708e+03+5.74706e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.26737e+03-3.01973e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 555,552...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.58157e+03-3.78946e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.46926e+03-2.95777e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.81937e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.46926e+03-2.95777e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.58157e+03+3.78946e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 556,552...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.49007e+02+1.15081e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.26703e+03+7.99373e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.22483e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.26703e+03+7.99373e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.49007e+02-1.15081e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 557,552...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.37497e+03+4.44834e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.32169e+01+1.10242e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.96537e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.32169e+01+1.10242e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.37497e+03-4.44834e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 558,552...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.36750e+03+9.56777e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.43129e+02+1.40633e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +8.30893e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.43129e+02+1.40633e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.36750e+03-9.56777e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 559,552...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.67628e+03-9.45409e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +6.20919e+02+2.36489e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.03166e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -6.20919e+02+2.36489e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.67628e+03+9.45409e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 560,552...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.45329e+03-2.25931e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.60118e+02-1.19522e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.00035e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.60118e+02-1.19522e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.45329e+03+2.25931e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 568,552...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.21760e+03+6.63018e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.79749e+03+7.12282e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.32186e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.79749e+03+7.12282e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.21760e+03-6.63018e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 533,553...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.95828e+02+5.94898e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -9.56197e+01+1.27960e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.35250e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +9.56197e+01+1.27960e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.95828e+02-5.94898e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 542,553...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +8.46793e+01+8.90211e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.06960e+03-1.17617e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.23720e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.06960e+03-1.17617e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +8.46793e+01-8.90211e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 549,553...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.26384e+03+3.29499e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.63199e+02-8.19230e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.76713e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.63199e+02-8.19230e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.26384e+03-3.29499e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 550,553...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.81762e+03+4.77543e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.30377e+03+2.77987e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +7.63104e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.30377e+03+2.77987e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.81762e+03-4.77543e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 551,553...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.45651e+03+1.77637e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.35179e+02+5.26420e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.44922e+02 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.35179e+02+5.26420e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.45651e+03-1.77637e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 552,553...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.08743e+02+3.45585e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.04971e+03+1.86039e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.28428e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.04971e+03+1.86039e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.08743e+02-3.45585e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 554,553...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.21812e+03-2.26116e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.13013e+03+2.88059e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.24429e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.13013e+03+2.88059e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.21812e+03+2.26116e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 555,553...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.61518e+03-4.33597e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.67097e+03+1.71882e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.42125e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.67097e+03+1.71882e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.61518e+03+4.33597e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 556,553...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.73474e+03+3.06702e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.65399e+03-1.30807e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.67344e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.65399e+03-1.30807e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.73474e+03-3.06702e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 557,553...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.78523e+02-4.41126e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -8.67287e+02-2.56778e+03i Hz
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```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.52802e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +8.67287e+02-2.56778e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.78523e+02+4.41126e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 558,553...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.27730e+03+1.42507e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +9.16816e+02-4.40761e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +6.02648e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -9.16816e+02-4.40761e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.27730e+03-1.42507e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 559,553...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.69696e+03-5.03901e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -3.30481e+03-4.60012e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.87804e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +3.30481e+03-4.60012e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.69696e+03+5.03901e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 551,554...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.34791e+02+3.73924e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.69507e+03-1.63506e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.44986e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.69507e+03-1.63506e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.34791e+02-3.73924e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 552,554...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.26737e+03+3.01973e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.80708e+03+5.74706e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.98762e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.80708e+03+5.74706e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.26737e+03-3.01973e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 553,554...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.1284 Hz
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[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 553,554...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.21812e+03-2.26116e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.13013e+03+2.88059e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.24429e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -3.13013e+03+2.88059e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.21812e+03+2.26116e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 555,554...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.12217e+03-1.60602e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.16748e+03-1.30823e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.82003e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.16748e+03-1.30823e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.12217e+03+1.60602e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 556,554...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.27739e+03+3.39467e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.09024e+03+1.46022e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.34190e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.09024e+03+1.46022e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.27739e+03-3.39467e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 557,554...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.61702e+03-3.73330e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -6.96329e+02-4.57237e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.25224e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +6.96329e+02-4.57237e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.61702e+03+3.73330e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 558,554...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.50496e+03-2.86485e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.20092e+03+1.98740e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.96203e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.20092e+03+1.98740e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.50496e+03+2.86485e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 559,554...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -6.66162e+02-1.26992e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +8.64309e+02+5.22618e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +8.80466e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -8.64309e+02+5.22618e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -6.66162e+02+1.26992e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 551,555...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.25853e+03+4.36876e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.98395e+03+1.49692e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.61522e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.98395e+03+1.49692e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.25853e+03-4.36876e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 552,555...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.58157e+03-3.78946e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.46926e+03-2.95777e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.81937e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.46926e+03-2.95777e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.58157e+03+3.78946e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 553,555...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
5.8057 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 553,555...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.61518e+03-4.33597e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.67097e+03+1.71882e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.42125e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -3.67097e+03+1.71882e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.61518e+03+4.33597e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 554,555...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 554,555...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.12217e+03-1.60602e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.16748e+03-1.30823e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.82003e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.16748e+03-1.30823e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.12217e+03+1.60602e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 556,555...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.61142e+04+2.64510e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.85489e+02-2.27515e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.32030e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.85489e+02-2.27515e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.61142e+04-2.64510e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 557,555...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -9.66208e+01-1.17894e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -9.39194e+02-8.65371e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.02948e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +9.39194e+02-8.65371e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -9.66208e+01+1.17894e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 558,555...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.12132e+03-1.56007e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.88846e+02-5.63331e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.48353e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.88846e+02-5.63331e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.12132e+03+1.56007e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 551,556...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.89983e+02+1.75024e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.73807e+02+4.66265e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.38280e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.73807e+02+4.66265e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.89983e+02-1.75024e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 552,556...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.49007e+02+1.15081e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.26703e+03+7.99373e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.22483e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.26703e+03+7.99373e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.49007e+02-1.15081e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 553,556...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
4.8952 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 553,556...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.73474e+03+3.06702e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.65399e+03-1.30807e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.67344e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -3.65399e+03-1.30807e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.73474e+03-3.06702e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 554,556...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 554,556...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.27739e+03+3.39467e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.09024e+03+1.46022e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.34190e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.09024e+03+1.46022e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.27739e+03-3.39467e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 555,556...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 555,556...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.61142e+04+2.64510e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.85489e+02-2.27515e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.32030e+04 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.85489e+02-2.27515e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.61142e+04-2.64510e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 557,556...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.75277e+03-7.82157e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.38096e+03-5.07138e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.06669e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.38096e+03-5.07138e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.75277e+03+7.82157e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 558,556...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.69042e+03-7.10349e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.35072e+03-1.85696e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.46019e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.35072e+03-1.85696e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.69042e+03+7.10349e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 559,556...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.75685e+03-9.99470e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.88850e+02-4.99589e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.64264e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.88850e+02-4.99589e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.75685e+03+9.99470e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 550,557...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -7.99367e+02+3.12766e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.45373e+03-1.90548e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.41523e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.45373e+03-1.90548e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -7.99367e+02-3.12766e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 551,557...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.71089e+03-4.16908e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.38723e+02-1.15522e+03i Hz
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```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.12396e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.38723e+02-1.15522e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.71089e+03+4.16908e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 552,557...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.37497e+03+4.44834e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -3.32169e+01+1.10242e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.96537e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +3.32169e+01+1.10242e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.37497e+03-4.44834e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 553,557...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
8.6445 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 553,557...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.78523e+02-4.41126e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -8.67287e+02-2.56778e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.52802e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +8.67287e+02-2.56778e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.78523e+02+4.41126e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 554,557...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.61702e+03-3.73330e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -6.96329e+02-4.57237e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.25224e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +6.96329e+02-4.57237e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.61702e+03+3.73330e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 555,557...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -9.66208e+01-1.17894e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -9.39194e+02-8.65371e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.02948e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +9.39194e+02-8.65371e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -9.66208e+01+1.17894e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 556,557...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.75277e+03-7.82157e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.38096e+03-5.07138e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.06669e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.38096e+03-5.07138e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.75277e+03+7.82157e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 558,557...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.52814e+02+7.89537e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.22538e+04+1.10015e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -7.45722e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.22538e+04+1.10015e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.52814e+02-7.89537e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 559,557...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.44254e+03-2.71357e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.24840e+04+5.55943e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.79023e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.24840e+04+5.55943e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.44254e+03+2.71357e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 550,558...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.40305e+02+1.64494e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.83253e+01+2.30652e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.37120e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.83253e+01+2.30652e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.40305e+02-1.64494e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 552,558...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.36750e+03+9.56777e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.43129e+02+1.40633e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +8.30893e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.43129e+02+1.40633e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.36750e+03-9.56777e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 553,558...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
4.5259 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 553,558...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.27730e+03+1.42507e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +9.16816e+02-4.40761e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +6.02648e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -9.16816e+02-4.40761e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.27730e+03-1.42507e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 554,558...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.50496e+03-2.86485e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.20092e+03+1.98740e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.96203e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.20092e+03+1.98740e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.50496e+03+2.86485e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 555,558...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.12132e+03-1.56007e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.88846e+02-5.63331e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.48353e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.88846e+02-5.63331e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.12132e+03+1.56007e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 556,558...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.69042e+03-7.10349e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.35072e+03-1.85696e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.46019e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.35072e+03-1.85696e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.69042e+03+7.10349e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 557,558...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 557,558...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.52814e+02+7.89537e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.22538e+04+1.10015e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -7.45722e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.22538e+04+1.10015e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.52814e+02-7.89537e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 559,558...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.36388e+04+8.53133e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.11167e+03-1.46704e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.24175e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.11167e+03-1.46704e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.36388e+04-8.53133e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 303,559...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +5.32700e+01-1.36965e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.38920e+02-5.60287e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +9.39156e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +5.38920e+02-5.60287e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +5.32700e+01+1.36965e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 550,559...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.84528e+03+6.65013e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.14682e+03+1.73804e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.66120e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.14682e+03+1.73804e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.84528e+03-6.65013e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 551,559...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.21275e+03-8.42476e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +9.74092e+01+3.10957e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.17633e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -9.74092e+01+3.10957e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.21275e+03+8.42476e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 552,559...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.67628e+03-9.45409e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +6.20919e+02+2.36489e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.03166e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -6.20919e+02+2.36489e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.67628e+03+9.45409e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 553,559...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
6.6846 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 553,559...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.69696e+03-5.03901e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.30481e+03-4.60012e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.87804e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.30481e+03-4.60012e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.69696e+03+5.03901e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 554,559...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -6.66162e+02-1.26992e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +8.64309e+02+5.22618e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +8.80466e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -8.64309e+02+5.22618e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -6.66162e+02+1.26992e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 556,559...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.75685e+03-9.99470e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.88850e+02-4.99589e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.64264e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.88850e+02-4.99589e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.75685e+03+9.99470e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 557,559...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
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[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 557,559...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.44254e+03-2.71357e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.24840e+04+5.55943e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.79023e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.24840e+04+5.55943e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.44254e+03+2.71357e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 558,559...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 558,559...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.36388e+04+8.53133e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.11167e+03-1.46704e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.24175e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.11167e+03-1.46704e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.36388e+04-8.53133e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 550,560...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.93466e+03+3.00462e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -8.23390e+03+3.46754e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.32949e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +8.23390e+03+3.46754e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.93466e+03-3.00462e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 552,560...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.45329e+03-2.25931e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.60118e+02-1.19522e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.00035e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.60118e+02-1.19522e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.45329e+03+2.25931e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 561,560...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.93512e+03+2.50801e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.95454e+02+5.29611e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.11856e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.95454e+02+5.29611e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.93512e+03-2.50801e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 562,560...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.84298e+03+7.59410e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -5.63813e+03+1.11609e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.13819e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +5.63813e+03+1.11609e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.84298e+03-7.59410e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 563,560...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.03046e+02+1.21302e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.46519e+03+1.14417e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.07561e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.46519e+03+1.14417e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.03046e+02-1.21302e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 564,560...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -7.39021e+03+1.67447e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.12341e+03-3.49422e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.65486e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.12341e+03-3.49422e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -7.39021e+03-1.67447e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 565,560...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.08342e+03-2.36903e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.01279e+03-5.73965e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.51172e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.01279e+03-5.73965e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.08342e+03+2.36903e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 568,560...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.63211e+02-1.49541e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +4.54042e+03+2.62167e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.08948e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -4.54042e+03+2.62167e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.63211e+02+1.49541e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 560,561...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
5.5593 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 560,561...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.93512e+03+2.50801e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.95454e+02+5.29611e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.11856e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.95454e+02+5.29611e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.93512e+03-2.50801e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 562,561...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.99193e+03-6.54738e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -4.92555e+02-3.07590e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.77351e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +4.92555e+02-3.07590e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.99193e+03+6.54738e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 563,561...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.07148e+02-2.16215e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -4.68224e+03-3.24994e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.86203e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +4.68224e+03-3.24994e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.07148e+02+2.16215e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 564,561...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.36818e+03-3.12733e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +8.57951e+02+1.72587e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.81637e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -8.57951e+02+1.72587e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.36818e+03+3.12733e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 565,561...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.22086e+02-1.63530e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -3.43310e+02-4.17524e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.28930e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +3.43310e+02-4.17524e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.22086e+02+1.63530e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 566,561...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.51712e+03-7.31470e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -9.81320e+02-1.39695e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.98722e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +9.81320e+02-1.39695e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.51712e+03+7.31470e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 567,561...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -6.37590e+03-1.19071e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.02877e+03+4.65539e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.69056e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.02877e+03+4.65539e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -6.37590e+03+1.19071e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 568,561...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.43991e+02+1.48622e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.41713e+03-1.28647e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.17611e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.41713e+03-1.28647e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.43991e+02-1.48622e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 560,562...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.84298e+03+7.59410e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -5.63813e+03+1.11609e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.13819e+03 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +5.63813e+03+1.11609e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.84298e+03-7.59410e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 561,562...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
12.8585 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 561,562...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.99193e+03-6.54738e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -4.92555e+02-3.07590e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.77351e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +4.92555e+02-3.07590e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.99193e+03+6.54738e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 563,562...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.71355e+03+1.45175e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.27414e+03+2.11982e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.32709e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.27414e+03+2.11982e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.71355e+03-1.45175e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 564,562...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
4.5463 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 564,562...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.08568e+02-3.22437e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.06191e+03+9.69667e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -7.10231e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.06191e+03+9.69667e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.08568e+02+3.22437e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 565,562...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
6.6066 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 565,562...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.58452e+03-2.26960e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +5.94044e+03+1.38993e+03i Hz
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```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.20643e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -5.94044e+03+1.38993e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.58452e+03+2.26960e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 566,562...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.70490e+02+1.64943e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.01939e+03-7.26063e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.05865e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.01939e+03-7.26063e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.70490e+02-1.64943e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 567,562...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.91631e+02+8.16016e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +9.77291e+02-1.23338e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.21683e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -9.77291e+02-1.23338e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.91631e+02-8.16016e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 568,562...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.85792e+03-1.63129e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.23144e+02+5.28070e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.17560e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.23144e+02+5.28070e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.85792e+03+1.63129e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 560,563...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -3.03046e+02+1.21302e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.46519e+03+1.14417e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.07561e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.46519e+03+1.14417e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -3.03046e+02-1.21302e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 561,563...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
2.9053 Hz
```

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[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 561,563...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.07148e+02-2.16215e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.68224e+03-3.24994e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.86203e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.68224e+03-3.24994e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.07148e+02+2.16215e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 562,563...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 562,563...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.71355e+03+1.45175e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.27414e+03+2.11982e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.32709e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.27414e+03+2.11982e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.71355e+03-1.45175e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 564,563...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
8.7339 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 564,563...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +8.38744e+02+9.12696e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.55158e+02+2.86015e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.28114e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.55158e+02+2.86015e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +8.38744e+02-9.12696e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 565,563...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
4.5441 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 565,563...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.06194e+03-3.15135e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +2.14369e+03+3.96442e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +8.73009e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -2.14369e+03+3.96442e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.06194e+03+3.15135e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 566,563...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.27714e+03+2.22232e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +6.62736e+03-1.61898e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -6.59285e+00 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -6.62736e+03-1.61898e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.27714e+03-2.22232e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 567,563...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.49053e+01+4.52342e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.79639e+03-1.49057e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.45749e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.79639e+03-1.49057e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.49053e+01-4.52342e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 568,563...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.54213e+03+1.06964e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.83065e+01+1.51132e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.53183e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.83065e+01+1.51132e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.54213e+03-1.06964e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 550,564...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.33581e+03+6.50393e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -8.86250e+02+2.04289e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +9.85807e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +8.86250e+02+2.04289e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.33581e+03-6.50393e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 560,564...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -7.39021e+03+1.67447e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.12341e+03-3.49422e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.65486e+03 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.12341e+03-3.49422e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -7.39021e+03-1.67447e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 561,564...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.36818e+03-3.12733e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +8.57951e+02+1.72587e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.81637e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -8.57951e+02+1.72587e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.36818e+03+3.12733e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 562,564...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.08568e+02-3.22437e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.06191e+03+9.69667e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -7.10231e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.06191e+03+9.69667e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.08568e+02+3.22437e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 563,564...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +8.38744e+02+9.12696e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.55158e+02+2.86015e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -3.28114e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.55158e+02+2.86015e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +8.38744e+02-9.12696e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 565,564...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.97949e+03-5.29343e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.54290e+04-5.84820e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -7.39756e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.54290e+04-5.84820e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.97949e+03+5.29343e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 566,564...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.93929e+01+2.66865e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.70179e+03+1.75317e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.26708e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.70179e+03+1.75317e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.93929e+01-2.66865e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 567,564...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.49114e+03+4.49490e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.37067e+02-8.13271e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.13830e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.37067e+02-8.13271e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.49114e+03-4.49490e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 560,565...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.08342e+03-2.36903e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.01279e+03-5.73965e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.51172e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.01279e+03-5.73965e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.08342e+03+2.36903e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 561,565...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +3.22086e+02-1.63530e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -3.43310e+02-4.17524e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.28930e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +3.43310e+02-4.17524e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +3.22086e+02+1.63530e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 562,565...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.58452e+03-2.26960e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +5.94044e+03+1.38993e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.20643e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -5.94044e+03+1.38993e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.58452e+03+2.26960e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 563,565...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.06194e+03-3.15135e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.14369e+03+3.96442e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +8.73009e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.14369e+03+3.96442e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.06194e+03+3.15135e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 564,565...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 564,565...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.97949e+03-5.29343e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.54290e+04-5.84820e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -7.39756e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.54290e+04-5.84820e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.97949e+03+5.29343e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 566,565...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.49512e+03+3.22468e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +4.00914e+00-1.24666e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.51697e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -4.00914e+00-1.24666e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.49512e+03-3.22468e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 567,565...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.48045e+03-4.16776e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.24285e+02-2.68839e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +8.72413e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.24285e+02-2.68839e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.48045e+03+4.16776e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 561,566...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.51712e+03-7.31470e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -9.81320e+02-1.39695e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.98722e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +9.81320e+02-1.39695e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.51712e+03+7.31470e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 562,566...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -5.70490e+02+1.64943e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.01939e+03-7.26063e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.05865e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.01939e+03-7.26063e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -5.70490e+02-1.64943e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 563,566...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.27714e+03+2.22232e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +6.62736e+03-1.61898e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -6.59285e+00 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -6.62736e+03-1.61898e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.27714e+03-2.22232e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 564,566...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.1061 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 564,566...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +7.93929e+01+2.66865e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.70179e+03+1.75317e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.26708e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.70179e+03+1.75317e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +7.93929e+01-2.66865e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 565,566...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
4.8305 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 565,566...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.49512e+03+3.22468e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +4.00914e+00-1.24666e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.51697e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -4.00914e+00-1.24666e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.49512e+03-3.22468e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 567,566...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.97752e+03-2.63661e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -1.51924e+04-3.77527e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.31625e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +1.51924e+04-3.77527e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.97752e+03+2.63661e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 561,567...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -6.37590e+03-1.19071e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +5.02877e+03+4.65539e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.69056e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -5.02877e+03+4.65539e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -6.37590e+03+1.19071e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 562,567...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.91631e+02+8.16016e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +9.77291e+02-1.23338e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.21683e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -9.77291e+02-1.23338e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.91631e+02-8.16016e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 563,567...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.49053e+01+4.52342e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.79639e+03-1.49057e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -4.45749e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.79639e+03-1.49057e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.49053e+01-4.52342e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 564,567...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
5.8494 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 564,567...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +4.49114e+03+4.49490e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.37067e+02-8.13271e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +5.13830e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.37067e+02-8.13271e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +4.49114e+03-4.49490e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 565,567...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
9.0981 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 565,567...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +2.48045e+03-4.16776e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.24285e+02-2.68839e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +8.72413e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.24285e+02-2.68839e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +2.48045e+03+4.16776e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 566,567...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 566,567...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.97752e+03-2.63661e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.51924e+04-3.77527e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.31625e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.51924e+04-3.77527e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.97752e+03+2.63661e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 550,568...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.60112e+02+4.33267e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -1.25730e+02+9.45966e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.10593e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +1.25730e+02+9.45966e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.60112e+02-4.33267e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 552,568...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.21760e+03+6.63018e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.79749e+03+7.12282e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.32186e+03 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.79749e+03+7.12282e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.21760e+03-6.63018e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 560,568...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -8.63211e+02-1.49541e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +4.54042e+03+2.62167e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -5.08948e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -4.54042e+03+2.62167e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -8.63211e+02+1.49541e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 561,568...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.43991e+02+1.48622e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.41713e+03-1.28647e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +2.17611e+02 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.41713e+03-1.28647e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.43991e+02-1.48622e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 562,568...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.85792e+03-1.63129e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.23144e+02+5.28070e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +4.17560e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.23144e+02+5.28070e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.85792e+03+1.63129e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 563,568...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.54213e+03+1.06964e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -4.83065e+01+1.51132e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.53183e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +4.83065e+01+1.51132e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.54213e+03-1.06964e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 569,568...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.80558e+03-8.26022e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +7.61106e+02+3.49320e+03i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.00684e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -7.61106e+02+3.49320e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.80558e+03+8.26022e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 570,568...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.30570e+03-1.41713e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +5.68558e+03+2.49245e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -6.59147e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -5.68558e+03+2.49245e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.30570e+03+1.41713e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 568,569...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
7.9442 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 568,569...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.80558e+03-8.26022e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +7.61106e+02+3.49320e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.00684e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -7.61106e+02+3.49320e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.80558e+03+8.26022e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 570,569...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.16673e+04+1.16447e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.86287e+02-6.92107e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.34453e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.86287e+02-6.92107e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.16673e+04-1.16447e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 571,569...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.95097e+03-4.00692e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.43986e+03+9.85703e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.56560e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.43986e+03+9.85703e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.95097e+03+4.00692e+02i Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 568,570...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
4.958 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 568,570...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -1.30570e+03-1.41713e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +5.68558e+03+2.49245e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -6.59147e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -5.68558e+03+2.49245e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -1.30570e+03+1.41713e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 569,570...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 569,570...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +1.16673e+04+1.16447e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.86287e+02-6.92107e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.34453e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.86287e+02-6.92107e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +1.16673e+04-1.16447e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 571,570...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.77563e+02-4.97966e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +1.65701e+03+1.56722e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.64639e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -1.65701e+03+1.56722e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.77563e+02+4.97966e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 569,571...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -4.95097e+03-4.00692e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +2.43986e+03+9.85703e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.56560e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -2.43986e+03+9.85703e+01i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -4.95097e+03+4.00692e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 570,571...
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.77563e+02-4.97966e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +1.65701e+03+1.56722e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +3.64639e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x -1.65701e+03+1.56722e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.77563e+02+4.97966e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 572,571...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.37535e+03+6.16624e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.26223e+03+5.44203e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.06762e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.26223e+03+5.44203e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.37535e+03-6.16624e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 573,571...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.13469e+01-1.51200e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -2.66212e+03-2.62480e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.53868e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +2.66212e+03-2.62480e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.13469e+01+1.51200e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 571,572...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
2.7239 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 571,572...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.37535e+03+6.16624e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x -4.26223e+03+5.44203e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -1.06762e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                   0.5*↙
(LzSm+LmSz) x +4.26223e+03+5.44203e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.37535e+03-6.16624e+02i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 573,572...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +9.29599e+03+1.25033e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                  -0.5*↙
(LzSp+LpSz) x +3.42041e+03-6.80518e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.12013e+04 Hz
```

```
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.42041e+03-6.80518e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +9.29599e+03-1.25033e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 571,573...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x↙
5.8435 Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 571,573...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x -2.13469e+01-1.51200e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x -2.66212e+03-2.62480e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x -2.53868e+03 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x +2.66212e+03-2.62480e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x -2.13469e+01+1.51200e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete isotropic coupling for↙
spins 572,573...
[...n > generator_liquid > relaxation > hamiltonian ]            (LxSx+LySy+LzSz) x -12↙
Hz
[...n > generator_liquid > relaxation > hamiltonian ]  complete anisotropic coupling for↙
spins 572,573...
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+2]:↙
0.5*(LpSp) x +9.29599e+03+1.25033e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,+1]:                 -0.5*↙
(LzSp+LpSz) x +3.42041e+03-6.80518e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2, 0]: sqrt(2/3)*(LzSz-0.25*↙
(LpSm+LmSp)) x +1.12013e+04 Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-1]:                  0.5*↙
(LzSm+LmSz) x -3.42041e+03-6.80518e+03i Hz
[...n > generator_liquid > relaxation > hamiltonian ]    T[2,-2]:↙
0.5*(LmSm) x +9.29599e+03-1.25033e+04i Hz
[...n > generator_liquid > relaxation > hamiltonian ]  55059 unique operators in the↙
Hamiltonian descriptor.
[...n > generator_liquid > relaxation > hamiltonian ]  building individual operators...
[...n > generator_liquid > relaxation > hamiltonian ]  operator construction took↙
41029.1915 seconds
[...n > generator_liquid > relaxation > hamiltonian ]  operator array footprint: 5.941 GB
[...n > generator_liquid > relaxation > hamiltonian ]  assembling the Hamiltonian...
[...n > generator_liquid > relaxation > hamiltonian ]  Hamiltonian assembly took 532.4441↙
seconds
[...n > generator_liquid > relaxation > hamiltonian ]  maximum spherical rank in the↙
giant spin Hamiltonian: 0
[...n > generator_liquid > relaxation > hamiltonian ]  memory footprint of H array:↙
0.054477 GB
[...n > generator_liquid > relaxation > hamiltonian ]  memory footprint of Q array:↙
1.9945 GB
[...noesy_ubiquitin > generator_liquid > relaxation ]  parallel evaluation path...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=+2, m=+2, p=+2,↙
q=+2, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=+2, m=+1, p=+2,↙
q=+1, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=+2, m=+0, p=+2,↙
q=+0, chemical species 1...
```

```
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=+2, m=-1, p=+2,↙
q=-1, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=+2, m=-2, p=+2,↙
q=-2, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=+1, m=+2, p=+1,↙
q=+2, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=+1, m=+1, p=+1,↙
q=+1, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=+1, m=+0, p=+1,↙
q=+0, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=+1, m=-1, p=+1,↙
q=-1, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=+1, m=-2, p=+1,↙
q=-2, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=+0, m=+2, p=+0,↙
q=+2, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=+0, m=+1, p=+0,↙
q=+1, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=+0, m=+0, p=+0,↙
q=+0, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=+0, m=-1, p=+0,↙
q=-1, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=+0, m=-2, p=+0,↙
q=-2, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=-1, m=+2, p=-1,↙
q=+2, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=-1, m=+1, p=-1,↙
q=+1, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=-1, m=+0, p=-1,↙
q=+0, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=-1, m=-1, p=-1,↙
q=-1, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=-1, m=-2, p=-1,↙
q=-2, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=-2, m=+2, p=-2,↙
q=+2, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=-2, m=+1, p=-2,↙
q=+1, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=-2, m=+0, p=-2,↙
q=+0, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=-2, m=-1, p=-2,↙
q=-1, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  queuing up L=2, k=-2, m=-2, p=-2,↙
q=-2, chemical species 1...
[...r_liquid > relaxation > redfield_integral_async ]  waiting for data from the nodes...
[...r_liquid > relaxation > redfield_integral_async ]  batch 1 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 2 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 3 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 4 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 5 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 6 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 7 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 8 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 9 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 10 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 11 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 12 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 13 out of 25 retrieved.
```

```
[...r_liquid > relaxation > redfield_integral_async ]  batch 14 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 15 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 16 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 17 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 18 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 19 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 20 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 21 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 22 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 23 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 24 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  batch 25 out of 25 retrieved.
[...r_liquid > relaxation > redfield_integral_async ]  Redfield superoperator done.
[...noesy_ubiquitin > generator_liquid > relaxation ]  full relaxation superoperator↙
density 0.013765%, nnz 112649709, sparsity 1
[...noesy_ubiquitin > generator_liquid > relaxation ]  dynamic frequency shifts have been↙
ignored.
[...noesy_ubiquitin > generator_liquid > relaxation ]  transverse cross-relaxation terms↙
have been ignored.
[...noesy_ubiquitin > generator_liquid > relaxation ]  thermalisation method: none,↙
relaxation to unit state
[...noesy_ubiquitin > generator_liquid > relaxation ]  final relaxation superoperator↙
density 0.0002305%, nnz(R)=1886331, sparsity 1
[...noesy_ubiquitin > generator_liquid > relaxation ]  memory footprint of the relaxation↙
superoperator: 0.034849 GB
[..._noesy_ubiquitin > generator_liquid > frqoffset ]  applying 4250 Hz Zeeman frequency↙
offset to 1H...
>>
```